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This instruction implements Department of Air Force Policy Directive (DAFPD) 21-1, *Maintenance of Military Materiel*, and establishes guidance consistent with Department of the Air Force Instruction (DAFI) 21-101, *Aircraft and Equipment Maintenance Management*, and Air Force Instruction (AFI) 21-131, *Joint Oil Analysis Program (JOAP).* This instruction provides guidance and direction necessary to develop an effective Aircraft Metals Technology (AMT) Program, Nondestructive Inspection (NDI) Program and Aircraft Structural Maintenance (ASM) Program. This publication is applicable to all Air National Guard (ANG) organizations (including ANG-led Active Associations and personnel during non-federalized periods that maintain aircraft, aircraft systems, equipment, support equipment, and components. It does not apply to Classic and AFRC-led ARC Associations. This publication does not apply to the Space National Guard. Ensure all records created as a result of processes prescribed in this publication are maintained in accordance with Air National Guard Manual (ANGMAN) 33-363, *Management of Air National Guard Records*, and disposed of in accordance with the Air Force Records Information Management (AFRIMS) Records Disposition Schedule located at https://www.my.af.mil/afrims/afrims/afrim/rims.cfm. Contact supporting records managers as required. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the DAF Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command to NGB/A4MM, 3501 Fetchet Avenue, Joint Base Andrews, MD 20762-5157. Supplements to this instruction will not lessen the requirements nor change the basic content or intent of this instruction. Process supplements in accordance with (IAW) DAFI 90-160, *Publications and Forms Management*. The authorities to waive wing level requirements in this

publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See Department of the Air Force Manual (DAFMAN) 90-161, *Publishing Processes and Procedures,* for description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate tier waiver approval authority, or alternately, to the Publication OPR for non-tiered compliance items using the DAF Form 679, *Department of the Air Force Publication Compliance Item Waiver Request/Approval*, to NGB/A4MM, 3501 Fetchet Avenue, Joint Base Andrews, MD 20762-5157. See **Attachment 1** for a glossary of references and supporting information. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Department of the Air Force.

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Chapter 1

AIRCRAFT METALS TECHNOLOGY PROGRAM (2A7X1)

1.1. MAJCOM/A4M Responsibilities.

1.1.1. The designated Senior Noncommissioned Officer (SNCO) or designee will manage the Aircraft Metals Technology (AMT) program and perform the following duties:

1.1.1.1. Establish base metal groups and positions required for welding certification in accordance with Technical Order (TO) 00-25-252, Intermediate Maintenance and Depot Level Maintenance Instructions - Aeronautical Equipment Welding.

1.1.1.2. Develop and manage command guidance and procedures for ANG AMT functions.

1.1.1.3. Approve all intra-command ANG AMT Temporary Duty (TDY) manning assistance requests.

1.1.1.4. Coordinates inter/intra-command 2A7X1 equipment transfers.

1.1.1.5. Forecast and ensure scheduling of 2A7X1 supplemental training.

1.1.1.6. Coordinate and approve Technical Order (TO) Publication Change Requests (PCR) and Source Maintenance and Recoverability Code reviews applicable to the AMT community.

1.1.1.7. Support the Air Force Metals Fabrication and Technology Office (MFTO) by participating in equipment evaluations, field surveys, Integrated Process Teams (IPT), Product Improvement Teams (PIT), managers' meetings/working groups and advisory board meetings.

1.1.1.8. Disseminate all official MFTO correspondence to the field. MFTO messages can be found at the following: https://usaf.dps.mil/teams/21761/SitePages/Home.aspx. Depending on the criticality of the MFTO message, maintenance cross-tells may be utilized to ensure official routing through Quality Assurance office to expedite compliance with any requirements outlined within the message.

1.1.1.9. Serve as the MAJCOM voting authority during the 2A7X1 Specialty Training Requirements Team (STRT) and Utilization and Training Workshop (U&TW). May appoint a 2A7X1 Subject Matter Expert (SME) to accompany them as an advisor to ensure proper advocacy for field-level requirements or as acting voting authority.

1.1.1.10. Review the MAJCOM Routine Inspection List (RIL) for 2A7X1 and submit changes to the MAJCOM Quality Assurance Functional Manager.

1.1.1.11. Identify and nominate a 2A7X1 (AMT) Single Point of Contact (SPOC) and Mission Design Series (MDS) AMT Subject Matter Experts (one for each MDS). Appointment made by ANG A4M Director or designee.

1.2. Maintenance Squadron Commander (MXS/CC) Responsibilities.

1.2.1. Ensure unit weld certification program is developed to meet MAJCOM and UTC requirements.

1.2.2. Ensure all 2A7X1 or civilian equivalent personnel requiring welder's certification or recertification to be certified at an Air Logistics Center (ALC) or equivalent to perform welding

operations when local certification capabilities do not exist, and no other Air Force Organization can support.

1.3. Fabrication Flight Superintendent Responsibilities.

1.3.1. Ensure all 2A7X1 journeyman, craftsman, and civilian equivalent welders assigned to the AMT section are certified in accordance with TO 00-25-252 to perform welding operation. Serves as certifying official for unit level welding examinations and executes all responsibilities IAW TO 00-25-252.

1.3.2. Ensure all full time Air National Guard (ANG) and civilian equivalent welders are certified IAW group and positions outlined by ANG Fabrication Carrer Field Functional Manager (CFFM).

1.3.3. Ensure all traditional Air National Guard (ANG) welders are certified IAW group and positions outlined by ANG Fabrication CFFM.

1.4. Aircraft Metals Technology (AMT) Section Chief Responsibilities.

1.4.1. Ensure machines and shop equipment are maintained and inspected IAW TO 34-1-3, Inspection and Maintenance - Machinery and Shop Equipment, and hand/measuring tools are maintained IAW TO 32-1-101, Use and Care of Hand Tools and Measuring Tools.

1.4.2. Ensure assigned AMT personnel maintain welding certifications outlined in paragraph 1.3.

1.4.3. Coordinate requests for an ALC or other qualified organization to qualify welders. If qualification and certifications are accomplished locally, coordinate certification requirements with the NDI section to ensure radiographic capability and required image quality indicators are present.

1.4.4. Ensure correct completion of DD Form 2757, Welding Examination Record, and AFTO Form 1, Welding Proficiency Log, for shop welders. DD Form 2757 can be found at: https://www.esd.whs.mil/Directives/forms/dd2500_2999/

1.4.4.1. The Observing Official is required to be a 5 or 7-level AMT technician or civilian equivalent welder.

1.4.4.2. The AMT Section Chief or designated representative will function as the testing official.

1.4.4.3. The visual inspector must be a 2A771 or civilian equivalent. The AMT section will perform visual and metallographic testing only. (T-2)

1.4.5. Ensure all military and non-ALC civilian personnel maintain proficiency by welding at least one ferrous and non-ferrous specimen in the Gas Tungsten Arc Welding (GTAW) process every 90 days and document on the AFTO Form 1 IAW TO 00-25-252.

1.4.6. Ensure journeymen are weld certified no later than (NLT) 24 months after award of 5-skill level.

1.5. Additive Manufacturing.

1.5.1. Local Purchase Equipment. Equipment items for Additive Manufacturing (AM) process will not be purchased locally without the knowledge of the MAJCOM Functional Manager and the Advanced Manufacturing Program Office (AMPO).

1.5.1.1. AM printers valued below $50K only require the immediate unit commander's written approval. AM printers valued above $50K require Air Force Life Cycle Management Center (AFLCMC) Rapid Sustainment Office (RSO) oversight and approval.

1.5.1.1.1. For purchase of AM equipment below $50,000, equipment makes, and model recommendations should be obtained from RSO-AMPO and Fabrication Career Field Functional Manager (NGB/A4MM).

1.5.1.2. Specific information on the application and requirements of AM technologies to include Supersonic Particle Disposition, also referred to as Cold-Spray (CS), can be requested through the RSO/AMPO using the org box outlined above.

1.5.1.3. Account for all AM printers, cold spray, and directed energy deposition machines in the Defense Property Accountability System (DPAS).

1.5.1.4. Consumables, support items, and replacement parts may be purchased at any time without approval.

1.5.1.4.1. Prior to local purchase of support items and replacement parts, consult the consumables list on the RSO-AMPO Teams page. This list contains materials and parts that are within the supply system. File path: General > Files > AF Filament NSN list.

1.5.2. Guidance for the use of AM to build replacement parts is prescribed in AFI 63-101/20101, Integrated Life Cycle Management.

1.5.3. Reference TO 34A-1-1, Additive Manufacturing Qualification of Technicians, Machines and Facilities, for AM methods and requirements for training, equipment, process controls, TO 34A-2-1, General Procedures and Process Controls Additive Manufacturing Metals, and TO 34A-3-1, Polymers Additive Manufacturing, General Procedures and Process Controls.

1.5.3.1. Technician qualification is not required per TO 34A-1-1, Additive Manufacturing Qualification of Technicians, Machines and Facilities, for the purposes of producing tooling, fixtures, jigs, training aids, prototype parts, or non-aircraft, non-aerospace equipment related items.

1.5.3.2. Formal training is only required for each process and type of equipment utilized to produce airworthy components.

1.5.4. Additive manufactured aircraft parts require approval from the applicable Weapon System Program Office (SPO).

1.5.4.1. Approved aircraft parts will be listed in the weapon system's illustrated parts breakdown (IPB) as an alternate AM part number and appropriate Source Maintenance, and Recoverability (SMR) code (i.e., MFO or MOO), reference TO 00-25-195, Source Maintenance, and Recoverability Coding of Air Force Weapons, Systems, and Equipment and Table 4-3., AF SMR Coding Matrix.

1.5.4.2. A Technical Assistance Request (TAR) or -107 will provide authorization for limited use of non-listed substitutes (supplies, components, support equipment, etc.) to prevent work stoppages, IAW TO 00-25-107, Maintenance Assistance.

1.5.4.3. The Joint Engineering Data Management Information Control System (JEDMICS) is the current repository for Technical Data Packages (TDPs). Accessing https://af-

jedmics.navair.navy.mil and entering the appropriate drawing number (i.e., 201874532) will download the complete TDP, including the embedded build file, required to additively manufacture the part.

1.5.4.4. For parts approved by -107, submit a Manufacturing Request Form, found on the RSO/AMPO Advanced Manufacturing Data Exchange Teams page. File path: General > Files > 02. AM Part, Part Repair & Engineering Request>Additive, to have a Technical Data Package (TDP) written and get the part added to JEDMICS.

1.6. CAD/CAM Software.

1.6.1. The recommended software for CAD is SolidWorks or Fusion 360.

1.6.2. Recommended CAM software: SolidCAM (native to SolidWorks 2018 and newer), Fusion manufacturing extension (Extension can be purchased as an add on to fusion 360), or MasterCAM.

1.7. Precision Cutting Tools.

1.7.1. Precision cutting tools (i.e., drill bits, reamers, taps, dies, end mills, etc.) are manufactured from high-speed steel (HSS), solid carbide or tool steel to achieve high hardness and wear resistance in accordance with TO 1-1A-9, Engineering Series for Aircraft Repair Aerospace Metals - General Data and Usage Factors. Manufacturer produced close tolerance cutting tools must adhere to National Aerospace Standards (NAS) (e.g., twist drills and reamers IAW Appendix C of TO 1-1A-1, Engineering Handbook Series for Aircraft Repair, General Manual for Structural Repair).

1.7.2. Manufacturer produced surface finishes are critical to the reduction of fatigue and stress. In accordance with TO 33B-1-1, Non-Destructive Inspection Methods, Basic Theory, any discontinuities marked into the tool (e.g., grinding, etching, stamping, etc.) will subject the tool to fatigue cracking and/or change tool run-out, resulting in premature tool failure, inaccurate airframe structure modifications, imprecise manufactured aircraft parts and/or damage to aircraft.

1.7.2.1. Unique or special circumstances apply to the management requirements of close tolerance, accurate, precision cutting tools, in accordance with DAFI 21-101, paragraphs 8.6.1.6 and 8.6.3.

1.7.2.2. Do not vibratory etch these tools as damage to the operator, tool, part, and/or aircraft may occur. Laser etching is appropriate; however, do not mark the shaft area of any arbor type tools as chuck slippage will remove the Equipment Identification Designator (EID). When laser etching is not practical, these tools will follow the marking and identification procedures in accordance with DAFI21-101_ANGSUP.

1.7.2.2.1. If a tool has been previously etched, do not remove the etching as this can further damage the tool.

1.7.2.3. Non-dispatchable industrial shop machinery accessories and attachments are managed in accordance with DAFI21-101_ANGSUP.

1.7.3. Precision measuring tools (i.e., calipers, micrometers, dial indicators, optical micrometers, 3D scanners, etc.) are manufactured with precision surfaces and precision layouts. Manufacturer

produced close tolerance measuring tools must adhere to National Institute of Standards and Technology (NIST) per TO 00-20-14, Air Force Metrology and Calibration Program.

1.7.3.1. Unique or special circumstances apply to the management requirements of precision measuring tools, in accordance with DAFI 21-101_ANGSUP.
1.7.3.2. Do not vibratory etch these tools as damage to the operator, or tool may occur. Laser etching is appropriate. When laser etching is not practical, these tools will follow the marking and identification procedures in accordance with DAFI21-101_ANGSUP.

1.7.3.2.1. Examples of tools covered by DAFI21-101_ANGSUP include clockwork mechanisms such as dial indicators and dial calipers, as well as tools that rely on optics like optical micrometers and 3D scanners.

1.7.3.2.2. If a tool has been previously etched, do not remove the etching as this can further damage the tool.

1.7.4. If the tool can be etched and has a protective surface coating, do not remove the protective coating to etch the tool.

1.8. Equipment Inspections.

1.8.1. The Defense Property Accountable System program (DPAS) Maintenance and Utilization (M&U) is the system of record for all aircraft and weapon system SE or SE Maintenance Information System (SEMIS).

1.8.2. Fabrication flights will follow TO 00-20-1, Aerospace Equipment Maintenance Inspection, Documentation, Policies, and Procedures and TO 34-1-3, Inspection and Maintenance - Machinery and Shop Equipment for machinery and industrial equipment inspections.

1.8.3. Sections will utilize DPAS M&U for inspection and maintenance requirements of machinery and industrial equipment unless the lack of full implementation precludes adoption of the Maintenance Information System (MIS). This includes machinery and industrial equipment with or without reoccurring inspection and maintenance requirements within their work center.

Chapter 2

NONDESTRUCTIVE INSPECTION PROGRAM (2A7X2)

2.1. MAJCOM/A4M Responsibilities.

2.1.1. The designated SNCO or designee will manage the NDI program and perform the following duties:

2.1.1.1. Manage the command NDI and Oil Analysis programs (OAP).

2.1.1.2. Develop and manage MAJCOM guidance and procedures for NDI and OAP functions.

2.1.1.3. Approve all intra-command NDI TDY manning assistance requests.

2.1.1.4. Coordinates inter/intra-command 2A7X2 equipment transfers.

2.1.1.5. Forecast and ensure scheduling of 2A7X2 supplemental training.

2.1.1.6. Coordinate and approve TO PCR and SMR code reviews applicable to the NDI community.

2.1.1.7. Support the Air Force NDI Office (AFNDIO) by participating in NDI equipment evaluations, field surveys, NDI IPT, NDI PIT, Air Force NDI managers' meetings/working groups, and advisory board meetings.

2.1.1.7.1. Disseminate all official AFNDIO messages to the field. AFNDIO messages can be found at the following: https://usaf.dps.mil/teams/22399/sitepages/home.aspx.

2.1.1.7.2. Depending on the criticality of the AFNDIO message, maintenance cross tells may be utilized to ensure official routing through the Quality Assurance office to expedite compliance with any requirements outlined in the message.

2.1.1.8. The Command Fabrication Superintendent will serve as the MAJCOM voting authority during the 2A7X2 STRT and U&TW. May appoint a 2A7X2 Subject Matter Expert (SME) to accompany them as an advisor to ensure proper advocacy for field-level requirements.

2.1.1.9. Review the NGB/A4MM Routine Inspection List (RIL) for 2A7X2 and submit changes to the NGB/A4MM Quality Assurance Functional Manager.

2.1.1.9.1. The RIL will include specific NDI tasks deemed critical for mission readiness. The NGB/A4MM Structures FAM will provide the NGB/A4MM QA FAM with an updated listing annually by August 1. (e.g., radiation safety, process control documentation, chemical concentration graphs, etc.).

2.1.1.9.2. A recommended RIL, special interest items (SII) listing, and supporting documents to aid non-NDI Quality Assurance (QA) Inspectors have been developed by AFNDIO and can be found at the following: https://usaf.dps.mil/teams/22399/sitepages/home.aspx.

2.1.1.10. Review NDI equipment evaluations and coordinate with AFNDIO to ensure accuracy during Program Objective Memorandum (POM) submissions.

2.1.2. Ensure civilian NDI technicians are National Aerospace Standard 410 (NAS 410) certified IAW DAFI 21-101, Aircraft and Equipment Maintenance Management and TO 33B1-1, Nondestructive Inspection Methods, Basic Theory.

2.1.3. Identify and nominate a 2A7X1 (NDI) Single Point of Contact (SPOC) and Mission Design Series (MDS) NDI Subject Matter Experts (one for each MDS). Appointment made by ANG A4M Director or designee.

2.2. Maintenance Squadron Commander (MXS/CC) Responsibilities.

2.2.1. Ensure only properly trained 2A7X2 personnel or NAS 410 certified civilian technicians operate NDI equipment and perform NDI IAW DAFI 21-101 and TO 33B-1-1.

2.2.2. Functional visits from the AFNDIO may be requested to assess the overall health of their NDI program and compliance with established standards.

2.2.3. Submit AFNDIO Functional visit requests through the MAJCOM NDI Functional Manager or designee to aflcmc-ezpt-ndio@us.af.mil using the approved request form located on the AF NDI Technical Community SharePoint: usaf.dps.mil/teams/22399/SiteAssets/Forms/AllItems.aspx?id=%2Fteams%2F22399%2FSiteAssets%2FLinkableFiles%2FAFNDIOAssessmentRequestForm_1Oct2023%2Epdf&parent=%2Fteams%2F22399%2FSiteAssets%2FLinkable Files

2.3. Fabrication Flight Superintendent Responsibilities.

2.3.1. Ensure capabilities exist to perform NDI methods applicable to the assigned weapon system(s) and support equipment (SE).

2.3.2. Communicate lab conditions and unsatisfactory technical proficiency issues to the MAJCOM NDI Functional Manager or designee when lab conditions render unsatisfactory NDI performance and require immediate corrective action. Report loss of NDI capabilities within 48 hours after the loss has been identified via email to the following: NGB.A4.A4MM-Policy@us.af.mil

2.3.3. Forecast funding for personnel to attend training courses and participate in applicable NDI Corporate Process Activities (CPA) and working groups.

2.4. Nondestructive Inspection (NDI) Section Chief Responsibilities.

2.4.1. Ensure machines and shop equipment are maintained and inspected in accordance with TO 33B-1-2, Non-Destructive Inspection - General Procedures and Process Controls.

2.4.2. Ensure NDI QA Augmentees are properly trained, documented and are current on all inspection methods IAW DAFI 21-101 and TO 33B-1-1.

2.4.2.1. Augmentees will conduct Personal Evaluations (PE) utilizing the PE Checklists located on the AFNDIO SharePoint.

2.4.2.2. Once the Evaluation is completed, augmentees are required to maintain these documents in the section. They will be maintained until the next PE is conducted on that specific method. (T-2)

2.4.3. Forecast funding for personnel to attend training courses and participate in applicable NDI Corporate Process Activities (CPA) and working groups.

2.4.4. Ensure the Radiation Safety Program requirements are compliant with the regulations TO 33B-11, Air Force Manual (AFMAN) 48-148, Ionizing Radiation Protection, and DAFMAN 48-125, Personnel Ionizing Radiation Dosimetry.

2.5. Precision measuring tools

2.5.1. Precision measuring tools (i.e., probes, transducers, optical equipment, etc.) are manufactured with precision surfaces and precision layouts. Manufacturer produced close tolerance measuring tools must adhere to National Institute of Standards and Technology (NIST) per TO 00-20-14, Air Force Metrology and Calibration Program.

2.5.2. Unique or special circumstances apply to the management requirements of precision measuring tools, in accordance with DAFI 21-101, paragraphs 8.6.1.6 and 8.6.3.

2.5.3. Do not vibratory etch these tools as damage to the operator, or tool may occur. Laser etching is appropriate. When laser etching is not practical, these tools will follow the marking and identification procedures in accordance with DAFI21-101, paragraph 8.3.6.6, 8.6.1.6, and 8.6.4.1.

2.5.3.1. If a tool has been previously etched, do not remove the etching as this can further damage the tool.

2.5.4. If the tool can be etched and has a protective surface coating, do not remove the protective coating to etch the tool.

2.6. Equipment Inspections.

2.6.1. The Defense Property Accountable System program (DPAS) Maintenance and Utilization (M&U) is the system of record for all aircraft and weapon system SE or SE Maintenance Information System (SEMIS).

2.6.2. Fabrication flights will follow TO 00-20-1, Aerospace Equipment Maintenance Inspection, Documentation, Policies, and Procedures and TO 34-1-3, Inspection and Maintenance - Machinery and Shop Equipment for machinery and industrial equipment inspections.

2.6.3. Sections will utilize DPAS M&U for inspection and maintenance requirements of machinery and industrial equipment unless the lack of full implementation precludes adoption of the MIS. This includes machinery and industrial equipment with or without reoccurring inspections and maintenance requirements within their work center. Exception: The NDI Process Control Program found in TO 33B-1-1, Section V & 33B-1-2 work package’s 102 through 106 does not fall within the scope of Aviation Support Equipment maintenance (AvSE). Process Control Automated Management System (PCAMS) will remain authorized at this time for all Process Control Documentation Methods per TO 33B-1-1, paragraph.

Chapter 3

AIRCRAFT STRUCTURAL MAINTENANCE AND CORROSION CONTROL PROGRAM (2A7X3)

3.1. MAJCOM/A4M Responsibilities.

3.1.1. The designated SNCO or designee will manage the Aircraft Structural Maintenance (ASM) and Corrosion Control programs and perform the following duties:

3.1.1.1. Develop and manage command guidance and procedures for ASM and Corrosion Control functions.

3.1.1.2. Approve all intra-command ASM TDY manning assistance requests.

3.1.1.3. Coordinates inter/intra-command 2A7X3 equipment transfers.

3.1.1.4. Forecast and ensure scheduling of 2A7X3 supplemental training.

3.1.1.5. Represent command at 2A7X3 utilization and training workshops. Provide corrosion and structural input to career field managers in all maintenance Air Force Specialty Codes (AFSCs).

3.1.1.6. Coordinate and approve TO PCR and SMR code reviews applicable to the ASM community.

3.1.1.7. Support the Air Force Corrosion Prevention and Control Office (AFCPCO) by participating in equipment evaluations, field surveys, Integrated Process Teams (IPT), Product Improvement Teams (PIT), managers' meetings/working groups and advisory board meetings.

3.1.1.7.1. Disseminate all official AFCPCO correspondence to the field. AFCPCO messages can be found at the following: https://dod.teams.microsoft.us/l/team/19%3adod%3acfcdeab8d43247d3906d3adc6cc7e60d%40thread.tacv2/conversations?groupId=d41fbdb9-4164-4bb5-8d9a-653e1290769b&tenantId=8331b18d-2d87-48ef-a35f-ac8818ebf9b4

3.1.1.7.2. Depending on the criticality of the AFCPCO message, maintenance cross-tells may be utilized to ensure official routing through Quality Assurance office to expedite compliance with any requirements outlined within the message.

3.1.1.7.3. Coordinate with the AFCPCO in selection and accomplishment of command Corrosion Survey at a minimum of every 5 years.

3.1.1.8. The Command Fabrication Superintendent will serve as the MAJCOM voting authority during the 2A7X3 STRT and U&TW. May appoint a 2A7X3 Subject Matter Expert (SME) to accompany them as an advisor to ensure proper advocacy for field-level requirements.

3.1.1.9. Review the NGB/A4MM Routine Inspection List (RIL) for 2A7X3 and submit changes to the NGB/A4MM Quality Assurance Functional Manager.

3.1.1.10. Serve as the Command Corrosion Control Manager.

3.1.1.10.1. Represent command at assigned weapon systems Corrosion Prevention Advisory Boards (CPAB), AF/DoD corrosion conferences, and field surveys. Advocate ANG maintenance unit attendance and active participation at weapon system specific CPABs.

3.1.1.10.2. Support Air Force Corrosion Control Prevention Executive (CCPE) by participating in working groups, advisory boards and providing corrosion data for the annual corrosion report.

3.1.1.10.3. Ensure adequate corrosion control training is available and current for all aircraft and Aerospace Ground Equipment (AGE) maintenance personnel.

3.1.1.11. Review MAJCOM paint schemes at a minimum of annually to ensure they meet mission requirements.

3.1.1.12. Identify and nominate a 2A7X3 (ASM) Single Point of Contact (SPOC) and Mission Design Series (MDS) ASM Subject Matter Experts (one for each MDS). Appointment made by ANG A4M Director or designee.

3.1.1.13. Manage command Low Observable (LO) programs

3.1.1.14. Coordinates inter/intra-command LO TDY manning assistance requests.

3.1.1.15. Develop and coordinate command policy and procedures for LO functions.

3.1.1.16. Coordinates inter/intra-command 2A7X5 equipment transfers.

3.1.1.17. Forecast and ensure scheduling of 2A7X5 supplemental training.

3.1.1.18. Represent NGB for all applicable 2A7X5 issues at LO conferences and meetings.

3.1.1.19. Serve as the NGB voting authority during the 2A7X5 Specialty Training Requirements Team (STRT) and Utilization and Training Workshop (U&TW).

3.2. Wing Commander (WG/CC) Responsibilities.

3.2.1. Approve all aircraft paint waivers, tail flashes, and nose art requests. Ensure all requests have been routed through local historian, Public Affairs, and wing corrosion manager. Submit all required documents to the HQ NGB/A4, Command Fabrication Superintendent/Corrosion Manager in accordance with paragraph 4.8.

3.2.2. Ensures funding is available to support annual radar cross section (RCS) flight test requirements.

3.3. Maintenance Group Commander (MXG/CC) Responsibilities.

3.3.1. Ensure adequate facilities, equipment, manpower, material, and funding are available to support a sound corrosion prevention and control program. The minimum requirements are:

3.3.1.1. Provide a year-round maintenance painting facility for assigned aircraft.

3.3.1.2. Facilities will meet Federal, State, and Local requirements. (T-2)

3.3.1.3. Ensure requirements outlined in DAFMAN 32-1084, Standard Facility Requirements, are met for SE and aircraft small parts. This capability can be incorporated in the aircraft corrosion control facility if space permits.

3.3.1.4. Ensure facility control technology meets local, state, and federal Environmental Protection Agency requirements in conjunction with current National Emission Standards for Hazardous Air Pollutants [Title 40, Code of Federal Regulations (CFR), Part 61 and 63].

3.3.2. Ensure adequate wash facilities are available on a year-round basis. This may be accomplished in any way deemed prudent for the locale and mission of the unit. This requirement may be met with one or more of the following:

3.3.2.1. A specially designed corrosion control facility completely enclosed, heated with environmentally controlled ventilation and waste disposal systems, and equipped with all utilities necessary for accomplishing all facets of aircraft corrosion control.

3.3.2.2. An environmentally compliant enclosed or covered wash rack.

3.3.2.3. An outside wash rack may be used on an interim basis when weather conditions permit and when approved by Base Civil Engineer.

3.3.3. Ensure frequency of wash/rinse cycles are maintained in accordance with TO 1-1-691, Aircraft Weapons Systems - Cleaning and Corrosion Control and Prevention, Aerospace and Non-Aerospace Equipment, and revised as necessary based on changes in mission and location.

3.3.3.1. For any aircraft overdue wash, request overfly approval from MAJCOM Corrosion Program Manager and Weapon Specific Systems Manager per TO 1-1-691, Aircraft Weapons Systems - Cleaning and Corrosion Control and Prevention, Aerospace and Non-Aerospace Equipment. Some airframes may require a submittal of a TAR/107.

3.3.3.2. Notification must include aircraft tail number(s), date of last wash, reason for overdue condition, and corrective action taken to prevent further occurrences. MAJCOM Corrosion Program Manager will ensure routing of waiver request to system program office (SPO) engineer and AFCPCO. The SPO engineer has final approval authority for waiver requests. (T-2)

3.3.4. Ensure Plans, Scheduling & Documentation section(s) schedule aircraft washes through applicable Maintenance Information System (MIS).

3.3.5. Ensure QA adequately evaluates corrosion control programs through inspection and maintenance follow-up evaluations.

3.3.6. Appoint a wing corrosion manager and aircraft wash facility manager, in writing, to provide continuity and ensure proper equipment and materials are maintained at the facility in accordance with DAFI 21-101.

3.3.7. Appoint personnel authorized to sign-off contract washes, as required.

3.3.8. Establish and maintain an effective low observables maintenance program.

3.3.9. Ensure scheduling of aircraft downtime for LO reduction at the appropriate time based on overall fleet health and/or SAS/LOHAS damage priority screen.

3.3.10. Appoint a fully qualified 2A775 technician or civilian equivalent to QA who is solely focused on LO maintenance processes. Requirement is 1 per AMU/FGS. LO QA personnel must possess a valid program security clearance to fully perform QA duties.

3.3.11. Ensure PS&D sections schedule F-22 and F-35 annual aircraft audits.

3.3.12. Monitor annual LO F-22 and F-35 audit trends to ensure fleet LO mission capable status is accurately documented and reported.

3.3.13. Support RCS test events with required aircraft and maintenance personnel. This includes Signature Management Program (SMP) flight testing, Acceptance Test Facility (turntable) revisits and Repair Verification Radar testing.

3.3.14. Units must confirm aircraft will support SMP at least 90 days prior to event.

3.3.15. Approval authority for flying aircraft in Aero Only configuration; ensures procedures contained in F-22A TOD and F-35 Interim Maintenance Procedure F35-IMP-A0110510105-960A-A are followed.

3.3.16. Work closely with the OG/CC to balance flying requirements with maintenance capability to prevent an uncontrollable LO backlog.

3.3.17. Ensure all personnel who directly perform maintenance on F-22 and F-35 aircraft complete annual LO awareness, panel handling training through the Maintenance Training Flight (MTF).

3.3.18. Ensure hot wash/lessons learned information is documented for each Theater Security Package (TSP) deployment and shared with like units/MAJCOMs.

3.4. Wing Corrosion Manager Responsibilities.

3.4.1. The wing corrosion program manager serves as the wing focal point for all aircraft and SE cleaning, corrosion and organic coatings related information and taskings. The wing corrosion program manager will organize, direct, and manage the wing/group corrosion management program according to: DAFI 21-101, DAFI 63-140, Aircraft Structural Integrity Program and Air and Space Equipment Structural Management, TO 1-1-691, TO 1-1-8, Application and Removal of Organic Coatings, Aerospace and Non-Aerospace Equipment, TO 1-1-689-3, Cleaning and Corrosion Control Volume III Avionics and Electrics, TO 35-1-3, Corrosion Prevention and Control, Cleaning; Painting, and Marking of USAF Support Equipment (SE), applicable weapon system specific –3 (structural repair manual), -23 (corrosion prevention and control manual), and this instruction. (T-2)

3.4.2. Before reassignment or retirement, the wing corrosion manager will ensure their successor is appointed early enough to provide an effective turnover of the corrosion program. The outgoing corrosion manager must confer with the Fabrication Flight Superintendent and ASM supervisors to identify a replacement. A copy of the new appointment memo will be sent to HQ NGBA4M, Command Fabrication Superintendent/Corrosion Manager, within 60 days of the appointment. (T-2)

3.4.3. Ensure corrosion inspections are accomplished during each phase/periodic/isochronal inspection for aircraft and equipment assigned, as required.

3.4.4. Ensure corrosion prevention and treatment procedures are performed within technical order requirements. In the event there are no weapons system specific post wash corrosion inspection requirements, the wing corrosion manager will coordinate with units to establish local requirements. (T-2)

3.4.5. Ensure only qualified product list (QPL) and/or the Qualified Product Database (QPD) authorized wash agents are utilized for overall and spot washes. Use of unapproved commercial or household/janitorial cleaners is strictly prohibited.

3.4.6. In conjunction with the local Supply/HazMat pharmacy, ensure only products from QPLs/QPDs approved for aircraft/aerospace equipment are being used.

3.4.7. Develop and submit comments or recommendations for improvement of the corrosion control program to HQ NGB/A4M, Command Fabrication Superintendent/Corrosion Manager.

3.4.8. Establish and chair a local corrosion prevention working group to formalize the wing corrosion management program. Working groups may meet as frequently as necessary to maintain an effective program but will meet at least annually. This working group should meet approximately 90 days prior to the next scheduled applicable weapons system CPAB to formalize action items. Minutes will be published and are recommended to be maintained at least 3 calendar years for continuity purposes. (T-2)

3.4.8.1. As a minimum, membership will include the unit corrosion manager, flight line (owning unit) maintenance supervisors, Plans Scheduling and Documentation (PS&D) personnel, ASM supervisors, AGE supervisors, and appropriate QA representatives. (T-2)

3.4.8.2. Submit CPAB action items to the Command Fabrication Superintendent/Corrosion Manager. Action items may be submitted throughout the year and must focus on structural integrity, extended service life, and improved repair techniques for the weapon system. (T-2)

3.4.8.3. Forecast funding, plan, and attend assigned weapon system CPAB or send a qualified representative.

3.4.9. Serve as wing corrosion program Point of Contact (POC) for all outside agencies.

3.4.10. Forecast or Program Objective Memoranda (POM) for funding requirements to attend DoD, Air Force and NGB Corrosion Manager meetings and workshops.

3.4.11. Ensure unit's corrosion related training courses are administered as intended by the MAJCOM and AFI. An initial interactive course with location specific supplemental training and annual refresher training is the minimum. See paragraph 3.15.

3.4.12. Determine the adequacy of corrosion control work cards for assigned equipment based on mission and location.

3.4.13. At units utilizing wash contractors, the wing corrosion manager must be thoroughly familiar with contract specifications, applicable technical orders, and inspection/acceptance criteria. The wing corrosion manager will be included in the coordination process of all new/updated wash contracts. (T-2)

3.4.14. Maintain records of all approved requests for aircraft names, nose art, tail flashes, internal nose art, score sheets of maintained aircraft, and wing corrosion manager appointment letter. Maintain full length color photographs of all approved aircraft names and nose art, along with approval documentation. All documents will be uploaded to the following: https://dod.teams.microsoft.us/l/team/19%3adod%3acfcdeab8d43247d3906d3adc6cc7e60d%40thread.tacv2/conversations?groupId=d41fbdb9-4164-4bb5-8d9a-653e1290769b&tenantId=8331b18d-2d87-48ef-a35f-ac8818ebf9b4

3.5. Fabrication Flight Superintendent Responsibilities.

3.5.1. Recommend a wing corrosion manager to the MXG/CC. (T-2) Forecast funding for wing corrosion control manager attendance at Corrosion Control Working Groups, CPABs, Aircraft

Structural Integrity Programs (ASIPs), and other pertinent meetings as required. Ensure Fabrication representation for ASIP and CPAB conferences in person or via telecom.

3.5.2. Recommend a qualified 2A753 or above as the wash rack facility manager to ensure proper cleaning materials, equipment and supplies are maintained in accordance with applicable technical orders, DAFI 21-101 and MAJCOM supplements. Not required when utilizing contracted washes as this position is captured in the contract.

3.6. Aircraft Structural Maintenance (ASM) Section Chief Responsibilities.

3.6.1. Ensure machines and shop equipment are maintained and inspected in accordance with TO 34-1-3, Inspection and Maintenance - Machinery and Shop Equipment and hand/measuring tools are maintained in accordance with TO 32-1-101, Use and Care of Hand Tools and Measuring Tools.

3.6.2. Recommend a wing corrosion manager to the Fabrication Flight Superintendent.

3.6.3. Ensure a corrosion control facility housekeeping program is developed and followed in accordance with DAFI 21-101.

3.6.4. Serve as the ASM technical assistant to the MXG/CCs and Command Fabrication Superintendent/Corrosion Manager.

3.6.5. Request depot assistance in accordance with TO 00-25-107, Maintenance Assistance, through the MAJCOM Weapon System Manager (WSM) with an information copy to NGB/A4M, Command Fabrication Superintendent/Corrosion Manager, when corrosion treatment/repairs exceed TO limits.

3.6.6. Oversee all aircraft specific Low Observable (LO) maintenance programs outlined by applicable MDS specific technical requirements and System Program Office (SPO) guidance.

3.6.7. Use LO Field Training Detachment (FTD) training availability/capacity at every opportunity to elevate capability over the long term and provide recommended changes as required.

3.6.8. Forecast funding to attend and participate in applicable LO meetings, CPABs and other structural related programs/meetings.

3.6.7. Ensure all personnel exposed to LO work environments comply with OSHA, AF, MAJCOM, and wing policies on hygiene standards and preventing contamination of common areas outside of the direct work environment.

3.6.8. Report fleet LO mission capable status (**Attachment 6**) to the applicable MAJCOM weapon system team daily. Fleet SAS/LOHAS average reported must not include non-possessed aircraft.

3.6.9. Maintain a comprehensive training plan ensuring assigned personnel develop and maintain proficiency in all facets of LO finishes, metallic structures, composite repair, corrosion control, signature assessment, and electronic maintenance information system data entry commensurate with awarded skill level.

3.6.10. Ensure LO ASM personnel receive pre-placement, special purpose, periodic and termination occupational physicals as deemed necessary by local Medical Group Aero Medical Services in accordance with AFI 48-145, *Occupational and Environmental Health Program.*

3.6.11. Ensure LO ASM personnel provide updated information required to obtain and retain special program security access to the unit or group security manager in a timely manner.

3.6.12. Ensure an LO composite repair facility security training plan is developed with initial and annual training to be used for each individual working within the section.

3.6.13. Develop a dedicated LO OML inspection crew to maintain inspection consistency and inspector proficiency.

3.6.14. Establish an OML team rotation plan to ensure all LO personnel remain proficient.

3.6.14. F-22s Request depot assistance in accordance with T.O. 00-25-107 through the ACC weapon system manager. F-35s Request depot assistance through ALIS utilizing the CRM process.

3.6.15. Ensure accuracy of LO mission capable status documentation and reporting **Attachment 4/Attachment 8.** This includes establishment of a SAS data integrity team to ensure data is routinely cross checked for accuracy.

3.6.16. Appoint an LO production supervisor to manage scheduling and workload, **Attachment 7**, on each shift.

3.6.17. Ensure no other maintenance is accomplished on the aircraft, equipment, or within the environmentally controlled/cordoned-off areas during corrosion prevention/treatment or coatings restoration when hazardous/toxic materials are in use requiring the use of specialized personal protective equipment.

3.6.18. Ensure deficiency reports (DR) are accomplished as necessary in accordance with T.O. 00-35D-54, *USAF Deficiency Reporting, Investigation, and Resolution*.

3.7. Aircraft Cleaning.

3.7.1. A complete exterior and interior cleaning will be accomplished on all aircraft in accordance with TO 1-1-691 and weapon system specific technical data. This will be accomplished during scheduled wash cycles, before isochronal or phase inspections, and prior to refurbishments. (T-2)

3.7.1.1. The following forms entries, as a minimum, are required for an aircraft wash:

3.7.1.1.1. “Aircraft wash required.” Enter this in the forms on a red dash. It is cleared by the owning unit aircraft wash supervisor.

3.7.1.1.2. “Aircraft taped and prepped for wash.” Enter this in the forms on a red X prior to the wash. It is cleared by the appropriate inspector after the aircraft has been de-taped, all associated equipment (such as wheel covers) is removed and associated tasks are accomplished, and the cleanliness inspection has been completed and signed off.

3.7.1.1.3. “Aircraft post-wash cleanliness inspection due.” Enter this in the forms on a red dash prior to the wash. It is cleared by the owning unit maintenance supervisor, production supervisor, or authorized contractor after completion of the cleanliness inspection. Note: Definition of clean: Surfaces must be deemed “clean” after satisfactory completion of the following method: Accomplish a close visual inspection to determine if all residue, oily film, and streaking has been removed. If cleanliness is questionable, a dry, lint free, white towel is wiped firmly across the

various surfaces. If excessive soiling of the towel occurs, the surface is not clean. Wheel wells, flap wells, and exterior surfaces should be inspected using this method. (T-2)

3.7.1.1.4. “Aircraft post-wash lubrication due.” Enter this in the forms on a red dash. It is cleared by the appropriate maintenance person responsible for ensuring task completion.

3.7.1.1.4.1. Proper post-wash lubrication is vital in prevention of corrosion. Lubrication prevents water intrusion in bearing cavities and subsequent corrosion damage.

3.7.1.1.4.2. If technicians wash components between normal cleaning cycles (flight line or “spot” washes), re-lubrication of the affected components is required.

3.7.1.2. If organizations know in advance that their aircraft or SE is scheduled to deploy, they must ensure aircraft and equipment washes are considered prior to mission deployment. If a wash was recently accomplished, the owning organization maintenance supervision will determine whether another wash is necessary prior to deployment. (T-2)

3.7.1.2.1. When an aircraft flies over salt water below 3,000 feet, the aircrew debriefing record and AFTO Form 781A, Maintenance Discrepancy and Work Document will be annotated with a “NOTE”. See TO 1-1-691 for complete guidance.

3.7.1.2.2. Aircraft properly rinsed in taxi-through, or “bird bath” type facilities, need not comply with this requirement. (T-2)

3.7.2. Aircraft latrine/urinal areas must be cleaned thoroughly to avoid corrosion damage due to effluent contamination. (T-2)

3.7.3. Interior areas will be dried after washing. Any method, such as low-pressure air, low temperature heat, or sponging/mopping, may be used. Standing water in any interior area of the aircraft must be removed. (T-2)

3.7.4. Pressurized water washing equipment, if authorized by the applicable system program office, may be used for aircraft washing in accordance with TO 1-1-691 and manufacturer’s instructions. However, all surfaces must be agitated with an authorized pad or other article. Pressure washing alone will not adequately remove contaminants from painted surfaces. (T-2)

3.7.4.1. Lubrication must be accomplished after all pressure washes in accordance with applicable technical data. (T-2)

3.7.4.2. All landing gear components will be hand washed and rinsed with low-pressure water. Refer to applicable landing gear technical orders for washing instructions. (T-2)

3.8. Maintenance Plans, Scheduling, and Documentation (PS&D).

3.8.1. Ensure frequency-of-cleaning/wash cycles in the MIS match those in TO 1-1-691. Monitors aircraft wash schedules to eliminate overdue washes.

3.8.2. Unit wash cycles will not exceed the maximum wash cycles listed in TO 1-1-691, unless coordinated and approved in accordance with TO 00-25-107. (T-2)

3.9. Aircraft Maintenance Unit Responsibilities.

3.9.1. Owning activities must wash and clean their aircraft and SE. (T-2) Wing Corrosion Manager and/or ASM personnel will assist the owning activities in their corrosion prevention

efforts by accomplishing scheduled corrosion inspections on aircraft, support, and test equipment. (T-2)

3.9.2. Only ASM personnel will perform aircraft inspection work cards specified for accomplishment by ASM in the -6 TO. All maintenance personnel, regardless of AFSC, must examine parts they remove for corrosion. (T-2)

3.9.3. Coordinate and schedule the use of wash rack facilities for other than isochronal/phase washes. For locations with contracted washes, aircraft maintenance responsibilities will be performed by the wash contractor in accordance with the wash contract. (T-2)

3.9.4. Appoint an experienced/qualified wash crew supervisor. This person will be trained according to paragraph 3.9.7. (T-2)

3.9.5. Ensure trained wash crew supervisors are present throughout the duration of aircraft washes.

3.9.6. Provide a task trained, appropriately equipped and qualified aircraft wash crew, to include as a minimum, a Dedicated Crew Chief (DCC) and/or Assistant Dedicated Crew Chief (ADCC) and personnel protective equipment within the work center.

3.9.7. The wing corrosion manager and owning unit supervisors/managers train and qualify personnel on aircraft washing and cleaning. Personnel assigned as wash supervisors, cleanliness inspectors, aircraft wash personnel and wash contractor quality assurance evaluators will complete the Aircraft Washing Procedures (Course C6ANU00TVT0001) video downloadable from https://367trss.cce.af.mil/Courses/Airframe?weaID=General. (T-2)

3.9.8. Ensure Aircraft Wash Supervisor and Employee's Certification is completed once during the initial wash training process and when work processes equipment, materials, or conditions change.

3.9.9. Ensure a cleanliness inspection of aircraft is accomplished after completion of the aircraft wash. An owning work center supervisor (production superintendent or dock chief, as appropriate) will sign-off the cleanliness inspection. (T-2) The key is to have supervisory personnel or production inspectors that did not participate in the wash perform the cleanliness inspection. Local requirements may be added to the checklist to enhance the unit cleanliness program.

3.9.9.1. The isochronal/phase inspection dock supervisor may accomplish the cleanliness inspection for isochronal/phase aircraft washes only.

3.9.9.2. Refer to paragraph 3.4.13 for contracted washes.

3.9.10. After the cleanliness inspection is completed, the inspector clears the AFTO Form 781A entry for "aircraft cleanliness inspection due after wash."

3.9.11. The wash supervisor ensures the facility and equipment is cleaned and properly stored at completion of each wash.

3.9.12. Maintenance personnel who remove/install aircraft panels and doors must ensure seals are serviceable and sealant applied to panels and fasteners as specified in applicable aircraft technical orders. (T-2)

3.9.13. Maintenance personnel must report all corrosion deficiencies through the applicable MIS in accordance with 00-20 series technical orders. Accurate documentation of maintenance actions in support of the corrosion control program is essential to support future manning, equipment requirements, training, and parts/material procurement requirements. (T-2)

3.9.13.1. Inspection, Crack/Corrosion and Repair Reporting (ICARR-3D) (C-130 users). NDI, ASM and QA personnel will use the ICARR-3D software to make inputs to the Automated Inspection, Repair, Corrosion, and Aircraft Tracking (AIRCAT) database for all NDI directed by technical orders; cracks and corrosion exceeding blending limits of Structural Repair Manual; and structural repairs in accordance with 1C-130A-6, Aircraft Scheduled Inspection and Maintenance Requirements/1C-130J-6, Airplane Scheduled Inspection. Corrosion within blending limits of the Structural Repair Manual will not be documented.

3.9.13.2. This is an Aircraft Structural Integrity Program (ASIP) requirement. See https://c130aircat.robins.af.mil/ for program instructions and information on ICARR-3D. Report all C-130 discrepancies in ICARR-3D. (T-2)

3.10. Wash Rack Facility Manager Responsibilities.

3.10.1. Ensure fall protection equipment is available, used and maintained in accordance with DAFMAN 91-203, Air Force Occupational Safety, Fire, and Health Standards, to allow coverage of all surface areas of aircraft during washing operations.

3.10.2. Ensure aircraft wash rack has qualified cleaners on hand as identified in weapon system specific technical data.

3.10.3. Ensure wash rack facility and surrounding area is kept clean and properly maintained.

3.10.4. Procure personal protective equipment used during wash process. Maintains wash rack facilities and equipment in serviceable condition (i.e., water hoses, pumps, air hoses, powered wash equipment, SE, Personal Protective Equipment (PPE), etc.). This may not apply to units utilizing wash contracts.

3.11. Wash Crew Supervisor Responsibilities.

3.11.1. Provide daily safety briefings explaining hazards associated with wash rack operations.

3.11.2. Ensure aircraft wash crews are task trained and qualified by reviewing the Aircraft Wash Procedures and Preventing Landing Gear Failure videos as a minimum (https://usaf.dps.mil/teams/21080/Corrosion/Training/Forms/AllItems) along with hands on training in accordance with TO 1-1-691. All training and qualifications must be documented in the personnel's training records. (T-2)

3.11.3. Ensure proper safety equipment, PPE and cleaning materials are serviceable and properly used in accordance with DAFMAN 91-203.

3.11.4. Enter the requirement for wash, signs the wash completion and enters the lubrication requirement in the AFTO Form 781A, Maintenance Discrepancy and Work Document, or other electronic form of documentation.

3.11.5. Ensure that fall protection is serviceable and inspected prior to use in accordance with DAFMAN 91-203.

3.11.6. Ensure aircraft are properly grounded as required in accordance with TO 00-25-172, Ground Servicing of Aircraft and Static Grounding/Bonding, and weapon system-specific technical data.

3.11.7. Inspect all wash rack equipment for serviceability (i.e. water hoses, pumps, air hoses, powered wash equipment, SE, etc.) prior to use.

3.11.8. Ensure wash rack facility, surrounding area and equipment is clean and equipment is properly stored before and after use.

3.12. Quality Assurance (QA) Responsibilities.

3.12.1. Evaluate at least 10% of all aircraft washes and at least 10% of all AGE washes for compliance with applicable technical data.

3.12.2. Evaluate the quality of 10% of all aircraft and equipment corrosion inspections.

3.12.3. Periodically review wash rack cleaning agents for QPL/QPD compliance.

3.12.4. Coordinate with the Wing Corrosion Manager to ensure an acceptance inspection is accomplished on all depots, Global Reach Improvement Program, and/or other off station paints upon return to home station. (T-2)

3.12.5. Ensure PPE is serviceable and properly utilized.

3.12.6. Contracting Officer Representative (COR) for aircraft washes will evaluate at least 10% of all aircraft washes. COR should maintain a file of discrepancies for consideration during contract rewrites. If a current contract specifies a different level of inspection than that specified herein, the contract will take precedence. Future contracts will incorporate the 10% inspection rate as a minimum. (T-2)

3.12.7. The COR will use locally developed aircraft wash cleanliness forms and checklists to evaluate contract wash compliance. (T-2)

3.12.8. Contract washes will be signed off by authorized personnel. (T-2)

3.12.9. LO QA personnel must possess a valid program security clearance to fully perform QA duties.

3.12.10. Establish a comprehensive inspection program to ensure the integrity of LO maintenance and SAS documentation associated with mission capable status reporting. This includes frequent personal evaluations of aircraft OML inspections, SAS documentation and LO repair processes in accordance with technical order guidance.

3.12.11. Identify/report all LO training and process deficiencies to the Fabrication Flight Chief.

3.12.12. Include periodic participation in annual aircraft LO audit inspections, **Attachment 5**, as part of Maintenance Standardization Evaluation Program.

3.12.13. Update Wing leadership on LO maintenance quality and potential improvement areas during the Health of Fleet (HoF) brief if applicable.

3.13. Aerospace Ground Equipment (AGE) Flight Production Responsibilities.

3.13.1. Ensure AGE work center personnel attend corrosion training.

3.13.2. The Wing Corrosion Manager, in concert with the AGE supervisor and unit maintenance training manager, will develop a corrosion prevention and control training curriculum. The AF Corrosion Prevention and Control Computer Based Training (CBT) is available on MyLearning. (T-2) The corrosion manager, in conjunction with the AGE supervisor, will determine the training interval. The training interval must be at least annually. (T-2)

3.13.3. Establish and enforce an effective corrosion program on assigned AGE and SE.

3.13.4. ASM and AGE supervisors determine repainting requirements.

3.13.4.1. AGE technicians will score the corrosion condition during periodic scheduled maintenance inspections utilizing the SE scoring in TO 35-1-3, Table 3-2 and annotate in DPAS.

3.13.4.2. Complete over coating of equipment is accomplished on an as needed basis. AGE will not be over coated solely for cosmetic purposes unless the AGE Flight and Fabrication Flight Superintendent determine it is required.

3.13.4.3. Complete over coating of equipment may be accomplished to apply the new equipment standard color (26173 FED-STD-595, MIL-PRF-85285). However, this must be accomplished on the units' regular corrosion schedule and equipment will be aligned with the new scheme on an attrition basis.

3.13.5. Owning work center personnel may treat small chips in the paint with Corrosion Prevention Compounds (CPC) listed in TO 35-1-3. For more permanent repairs of small, chipped areas, use authorized coating systems that are contained in items such as but not limited to: Sempens, Preval compressed air spray packs, Clip-Pacs, Brush and Roller, or Akzo Nobel's Spray 2 Fix aerosol can. Larger areas will be treated by the aircraft structural maintenance work center or, if applicable, contracted sources. (T-2)

3.13.5.1. Units will familiarize themselves with AGE painting materials and processes in accordance with TO 35-1-3 prior to awarding off-base contracts to get AGE painted.

3.13.5.2. Units will verify specifications for primer and topcoat, and color number requirements and ensure that these are addressed in the contract. (T-2)

3.13.6. AGE SE will be painted in accordance with TO 35-1-3. (T-2)

3.13.7. Ensure an automated system is used to schedule and document AGE painting. A historical entry will be made into the automated system upon complete repainting of equipment. (T-2)

3.13.8. Enforce the proper use of approved cleaning compounds in accordance with TO 35-1-3 and the QPL or QPD.

3.14. Aerospace Coating and Paint Score Requirements.

3.14.1. Coating System Scoring and Maintenance. All units will score aircraft coating systems to determine frequency of topcoat application. (T-2)

3.14.2. Scoring will be accomplished as required during each Isochronal Inspection (ISO), Home Station Check, A-Check, all transfers, and upon return from depot maintenance. (T-2)

3.14.3. The exterior of aircraft must be clean prior to paint scoring. Supervisors will use ratings to determine corrosion treatment/paint scheduling priority. (T-2)

3.14.4. Units will adopt maintenance-painting techniques (i.e., spot painting and sectionalized painting as stated in TO 1-1-8) to maintain aircraft corrosion protection and appearance between overcoats.

3.14.5. A locally developed tracking system will be utilized to annotate scores. Score sheets will be uploaded to the NGB/A4M Corrosion Prevention & Control team at: https://usaf.dps.mil/sites/ngba4/SitePages/A4MM---Fabrication.aspx

3.15. Corrosion Prevention and Control Training.

3.15.1. All aircraft maintenance personnel will receive general corrosion prevention and identification refresher training at least annually, they will also receive local and unique corrosion awareness training. Ensure sufficient training opportunities are provided for classic associate unit personnel during Unit Training Assembly days. Training will be a combination of Interactive Multimedia Instruction (IMI) and local and unique corrosion awareness training will be developed by the Wing Corrosion Manager.

3.15.1.1. IMI Training will consist of the Corrosion Control Familiarization Course 1 downloadable from https://www.youtube.com/watch?v=EHMW0_iKzPs. Video can also be found on the Air Force Corrosion Prevention and Control Office (AFCPCO) website at https://www.my.af.mil/gcssaf/USAF/ep/browse.do?programId=t88B4F00B441D422B014427477A10019B&channelPageId=s6925EC133EFE0FB5E044080020E329A9.

3.15.1.2. AFSC 2A7X3 (ASM) personnel and/or equivalent are exempt from periodic corrosion familiarization training.

3.15.2. If group block training method is used, supplemental training is conducted by the Wing Corrosion Manager or designated representative holding a primary AFSC of 2A773 or 2A790. If block or refresher training is done on an individual basis, the supplemental training should be self-supporting, such as a short video, PowerPoint presentation, or other medium that the individual can review.

3.15.3. The Wing Corrosion Manager, in conjunction with the unit maintenance training manager, develops formal classroom training curriculum. As a minimum, the curriculum will include: (T-2)

3.15.3.1. Corrosion identification procedures and techniques using the most current available Air Force aircraft corrosion visual training aids and information.

3.15.3.2. Identification of corrosion prone areas on unit specific weapon systems and equipment.

3.15.3.3. Reporting and documentation procedures for identified corrosion.

3.15.3.4. Importance of proper selection and use of sealants, CPC, and lubricants.

3.15.3.5. Proper selection and use of all cleaning materials.

3.15.4. The Wing Corrosion Manager periodically updates training material and information with the assistance of the unit maintenance training manager and information gained from CPABs and corrosion manager's conferences.

3.15.5. Periodic corrosion training does not replace normal on-the-job training requirements in any career field.

3.16. Equipment Inspections.

3.16.1. The Defense Property Accountable System program (DPAS) Maintenance and Utilization (M&U) is the system of record for all aircraft and weapon system SE or SE Maintenance Information System (SEMIS).

3.16.2. Fabrication flights will follow TO 00-20-1, Aerospace Equipment Maintenance Inspection, Documentation, Policies, and Procedures and TO 34-1-3, Inspection and Maintenance - Machinery and Shop Equipment for machinery and industrial equipment inspections.

3.16.3. Sections will utilize DPAS M&U for inspection and maintenance requirements of machinery and industrial equipment unless the lack of full implementation precludes adoption of the MIS. This includes machinery and industrial equipment with or without reoccurring inspections and maintenance requirements within their work center.

3.17. Precision measuring tools.

3.17.1. Precision measuring tools or ASM shall follow the same criteria as described in **Chapter 1.**

3.18. Low Observable/Composite Repair Facility (LO/CRF) Manager Responsibilities

3.18.1. Reporting facility operation deficiencies such as the Heating, Ventilation and Air Conditioning (HVAC) systems, compressed and breathing air systems, electrical systems, plumbing and drainage systems within the LO/CRF and on assigned real property of the LO/CRF.

3.18.2. Perform tasks related to the overall management and operations of the LO/CRF, including energy management and equipment inventory.

3.18.3. Inspect facility for safety and maintenance issues that require repair as required by technical orders and Air Force instructions. Ensure all measures are taken to maintain security accreditation of facility as required.

3.18.4. Submit facility work orders through Civil Engineer (CE) Customer Service. In cases of established/approved contract maintenance for facilities, contact contractor for emergency or out of cycle maintenance. This includes warranty repairs and required maintenance.

3.18.4.1. Track the completion of contractor and CE work orders. Maintain record of all work to include response time and time required until satisfactory completion of work.

3.18.4.2. Perform escort duties or provide escorts as needed for contract maintenance personnel within special access areas of the LO/CRF.

3.19. LO Support Responsibilities.

3.19.1. Stocks supplies, consumables, tools, and equipment to support LO aircraft inspection, maintenance, and surface treatment. Processes supply requests, tracks MICAP due-outs, monitors bench stock, conducts bench stock and adjusted stock level reviews, and operates tool storage areas. In addition, the Support Section NCOIC will ensure a section due-out release point and holding bins are established. Supports CTK/Special tools, E- tools and test equipment.

3.19.2. Maintain bench and operating stocks.

3.19.3. Control and maintain TMDE in accordance with T.O. 33-1-27, *Maintenance Support of Precision Measurement Equipment*, T.O. 33K-1-100-2-CD-1, *TMDE Calibration Interval Technical Order and Work Unit Code Reference Guide*; T.O. 00-20-14, *Air Force Metrology and Calibration Program* and other applicable technical directives concerning the use, care, handling, transportation, and calibration of TMDE owned by the section.

3.19.4. Provide monthly critical support equipment status update to squadron supervision.

3.19.5. Maintain QRL as needed and provide it to technicians.

3.19.6. Track and process DIFM assets.

3.19.7. Manage reusable containers in accordance with DAFI 24-602 Volume 2, *Cargo Movement*, and T.O. 00-20-3, *Maintenance Processing of Reparable Property and the Repair Cycle Asset Control System*.

3.19.8. Manage section's hazardous material (HAZMAT) program.

3.19.8.1. Organize and coordinate effective shipping and staging operation.

3.19.8.2. Issue/turn in materials to/from workers using the tool accountability system (TAS).

3.19.8.3. Maintain records and documentation actions to ensure compliance with applicable directives.

Chapter 4

AIRCRAFT MARKING GUIDANCE

4.1. Paint Schemes/Configurations and USAF Standard Markings.

4.1.1. Paint schemes/configurations and USAF standard markings will be applied in accordance with the weapon system specific aircraft TO and aircraft drawings, TO 1-1-8 and this instruction.

When a conflict exists between technical orders and instructions, the weapon system specific TO and aircraft drawings will take precedence.

4.2. Exterior Markings / Coatings.

4.2.1. All aircraft markings will be maintained intact, legible, and distinct in color (not faded). Command standardization of markings by Mission Design Series (MDS) is of primary concern. (T-2)

4.2.2. All exterior aircraft markings must match the gloss level of the basecoat. No approved diffuse clear coats are available; low-gloss materials must be used for all markings on aircraft with lusterless paint schemes. (T-2)

4.2.3. Operational markings and structural coating/corrosion maintenance will take precedence over cosmetic refinements. (T-2)

4.2.4. When large sections of an aircraft are painted (i.e., entire wing, fuselage, or empennage), they will be documented in applicable MIS and the individual AFTO Form 95, Significant Historical Data. (T-2)

4.2.5. Review applicable weapon system technical data for Weight and Balance (W&B) requirements.

4.3. Interior Markings.

4.3.1. Aircraft interior markings are permitted to be displayed within the aircraft, including unit, wing, MAJCOM identifiers, aircraft tail/serial number, tail flash, DCC/ADCC, and U.S. Air Force markings.

4.3.2. Markings must not be visible from the aircraft's exterior when access/entry doors and/or cargo-passenger ramp is open.

4.3.3. Markings will be uniform in nature (i.e., all interior markings must be placed in a standardized location for all aircraft of the same MDS).

4.3.4. All levels of supervision have the responsibility to review the markings for tastefulness, appropriateness, and adherence to copyright laws.

4.4. Aircraft Mandatory Markings.

4.4.1. Mandatory Markings:

4.4.1.1. United States Flag.

4.4.1.2. National Star Insignia.

4.4.2. Mandatory Marking Application. Mandatory markings will be applied in accordance with applicable weapon system specific TO and drawings, TO 1-1-8, and the applicable table in this instruction. Table A2.1 through Table A2.6 in Attachment 2 lists the size, location, and color of markings by aircraft type. For identification, placement, and color of mandatory markings other than those identified in this instruction, refer to the weapon system specific TO and drawings. (T-2)

4.4.3. US Flag. Paint may be used only when high-quality templates or silk-screen processes are used. Flag decals can be obtained by going online to the Defense Logistics Agency (DLA)

Document Services website at https://www.dla.mil/Document-Services/. Customer support may be reached at 1-866-736-7010. Flag decals may be purchased with the International Merchant Purchasing Authorization Card. There is no form number or part number for flag decals; therefore, a "short title" should be used. The short title is either "21-inch by 40-inch Matte Finish Flag Decal" or "24-inch by 48-inch Matte Finish Flag Decal," as applicable. Flag decals have a one-year shelf life. For best results, use 3M edge sealer part #4150 (designed for polyester decal films). The National Star Insignia will be applied in accordance with applicable weapon system specific TO and drawings, TO 1-1-8, and the applicable table in this instruction.

4.5. Heritage Aircraft Markings.

4.5.1. Heritage Aircraft. Each Wing is authorized the designation of one "heritage" aircraft per Mission Design Series (MDS) to be the wing pride aircraft; this aircraft is authorized the application of the legacy markings stated below. Table A3.1 through Table A3.6 in Attachment 3 lists the size, location, and color of markings by aircraft type. Note: All exterior markings not identified in paragraph 4.4 are considered heritage markings.

4.5.1.1. Units are not required to remove current heritage markings. Removal of these markings will occur through the normal programmed depot maintenance of each assigned aircraft.

4.5.1.2. The Wing Corrosion Manager or designee will identify all heritage aircraft for each wing and submit to NGB.A4.A4MM-Policy@us.af.mil for record-keeping. Should the selected heritage aircraft change at any point in time, the Wing Corrosion Manager or designee will resubmit the required information to identify the newly selected aircraft. Submissions must contain the following:

4.5.1.2.1. MDS and aircraft Serial Number.

4.5.1.2.2. Date identified as a heritage aircraft.

4.5.1.2.3. Digital color photographs, one of each side of aircraft. The use of a slide presentation format is allowed.

4.5.1.3. Nose Art. Nose art is authorized for all ANG owned aircraft. LO aircraft will use coating materials approved for that specific aircraft. No vinyl or decals will be allowed on the outer mod line of an LO aircraft. Wing Commanders must approve all nose art, in writing, and be responsible for issues associated with its application. Nose art will not exceed 18 inches for fighter aircraft, 36 inches for large aircraft, not infringe on mandatory aircraft markings and meet the gloss requirements of the original paint scheme. Units that are called upon to perform in a hostile environment may be required to remove nose art prior to deployment. All nose art applied to wing aircraft will be of standard size and location. Nose art and tail flash design requests must follow the coordination process outlined in paragraph 4.5.1.3.4 for approval prior to installation. (T-2)

4.5.1.3.1. Wing mascot/logo may be applied as nose art.

4.5.1.3.2. On aircraft with lusterless paint schemes, nose art must be applied using lusterless paint and/or decals. (T-2)

4.5.1.3.3. DELETE.

4.5.1.3.4. Units will forward a clear/detailed color photograph of their aircraft that depict nonstandard markings. A letter of approval from the Wing Commander must accompany all photos. Photos (digital) will be provided to NGB/A4MM for approval. If markings are changed in the future, new photos and approval letter from the Wing Commander must be forwarded within 30 days of the change. Original letters of approval must be kept on file at the unit for inspection purposes. Digital photos can be sent by email (call DSN 612-8488 for email address) or provided on CD by mail to: NGB/A4MM, Attention: Maintenance Functional Manager, 3501 Fetchet Avenue, Andrews AFB MD 20762-5157.

4.5.1.4. Temporary Nose Art. Award ceremony aircraft displays with temporary nose arts (Knuckle Buster, Wing awards, etc.) must follow the coordination process outlined in paragraph 4.5.1.3.4 for approval. Aircraft with temporary markings will not fly until temporary markings have been removed. The duration of time an aircraft may display any temporary nose art will be determined by the Wing Commander. Note: This does not apply to aircraft that are identified as a heritage aircraft. (T-2)

4.5.1.5. Internal Nose Art. Requestors will follow the coordination process outlined in paragraph

4.5.1.3.4 for approval. No alterations to the aircraft frame will be made to accommodate internal nose art. Examples of approved internal nose art consist of flags, family photos, paintings, rugs, latrine paint schemes, crew placards with nicknames. Exception: paragraph 4.5.1.13.3 still stands as no external nicknames allowed. (T-2)

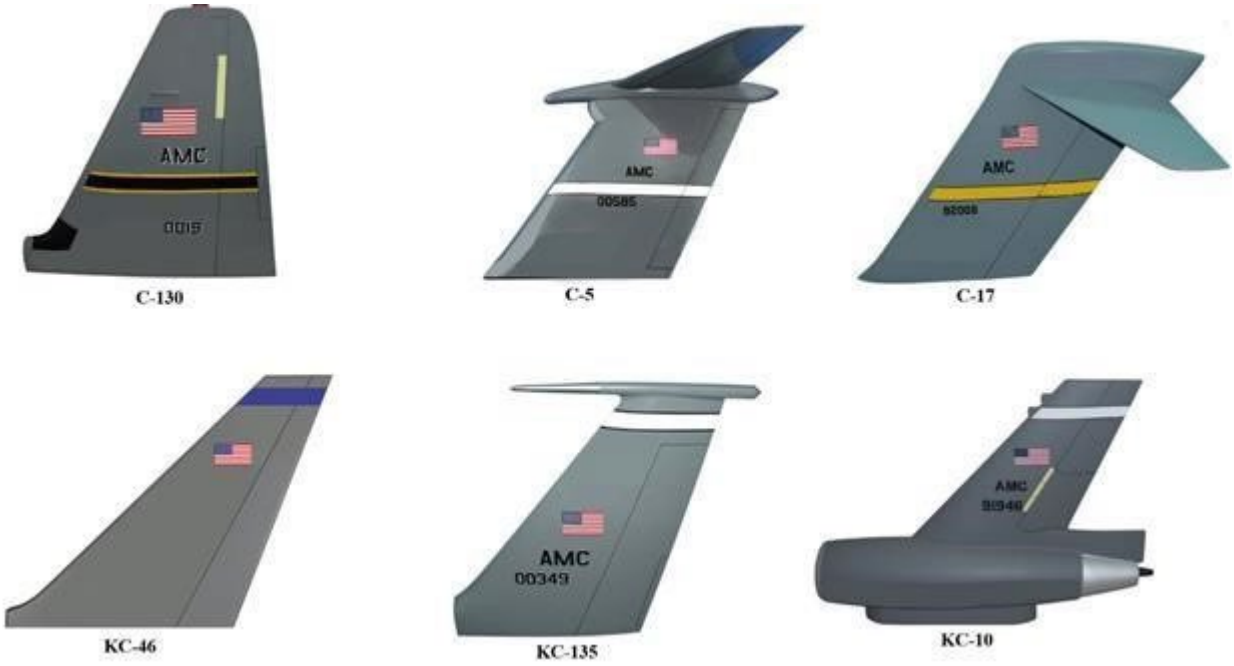
4.5.1.6. Tail Flash. Requestors must follow the coordination process outlined in paragraph 4.5.1.3.4 for approval.

4.5.1.6.1. May contain colors/numbers of all squadrons assigned to the wing, however, must remain within the tail band stripes specified in the applicable table. Refer to section 4.2. (T-2)

4.5.1.6.2. Wing/Squadron/Aircraft Maintenance Unit Colors. Each operational squadron may have its colors and/or logos applied within the boundaries of the tail stripes, or the entire wing may share one tail stripe design.

4.5.1.6.3. On aircraft with lusterless paint schemes, tail flash must be applied using lusterless paint and/or decals. (T-2)

Figure 4.1. Heritage Tail Flash Configurations.

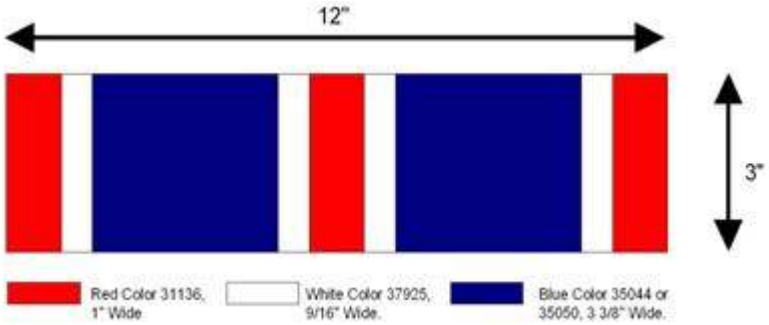


4.5.1.6.4. Letters and Numerals. These markings may be applied to heritage aircraft only using any style letter/numeral (font) deemed appropriate by the WG/CC. Size and location must remain standardized for all wing-assigned heritage aircraft. (T-2)

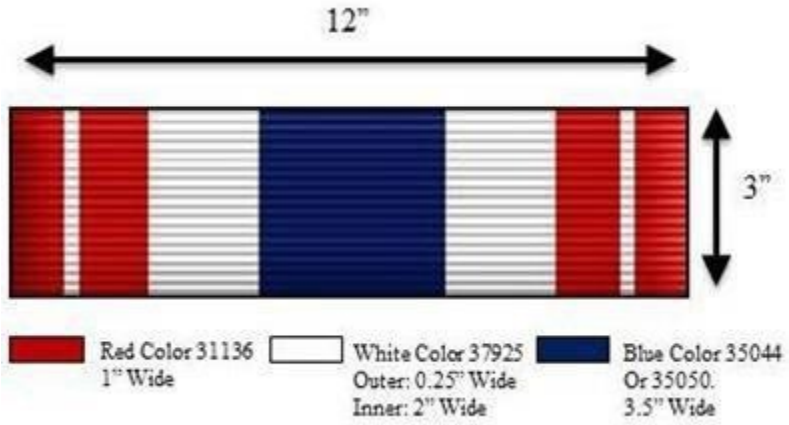
4.5.1.6.5. Aircraft Names. Aircraft Names are authorized and may be applied to heritage aircraft only. The proposed name must either have a national or military theme or honor a locale adjacent to an ANG base or aircraft manufacturing point. Route recommendations through your WG/CC to include the proposed name and detailed justification. If applied in addition to nose art, the aircraft name and nose art must be complementary; the font, size, and location may be changed to complement the nose art. (T-2)

4.5.1.6.6. Air Force Outstanding Unit Award. The Air Force Outstanding Unit Award (AFOUA) may be applied to heritage aircraft only, if applicable. AFOUA decals, with and without oak leaf clusters, are available from https://www.dla.mil/Document-Services/.

Figure 4.2. Outstanding Unit Award.



4.5.1.6.7. Air Force Meritorious Unit Award (MUA) may be applied immediately adjacent to the AFOUA on heritage aircraft only, if applicable. Figure 4.3. Meritorious Unit Award.



4.5.1.6.8. Boom Elevator Markings. Boom elevator markings may be applied to heritage aircraft only, in accordance with the appropriate table, weapon system specific TO, and/or weapon system drawings with the approval of the WG/CC.

4.5.1.6.9. WG/CC or WG/CV names and Group Commander's names (list all group commanders or none) may be used in place of the DCC names on heritage aircraft only. The wing designator may be included in the name block. Prior to deployment or flight into a combat zone (including transient aircraft), all names will be removed from the aircraft. (T-2)

4.5.1.6.10. DCC/ADCC. If elected, DCC and ADCC names will be applied in accordance with this instruction and placed on interior placards. Exterior application may be applied to heritage aircraft only. The preferred method is to apply and remove decals utilizing appropriate vinyl. This will allow all units to participate in the DCC program. See Attachment 3, Table A3.1 through Table A3.6. Units will be consistent when selecting interior or exterior placards. Prior to deployment or flight into a combat zone (including transient aircraft), all names will be removed from the aircraft. (T-2)

4.5.1.6.11. Transient aircraft, as defined in TO 00-20-1, are not deployed and are not subject to the restrictions in paragraphs 4.5.1.12 and 4.5.1.13 so long as flights do not transition through a combat zone.

4.5.1.6.12. The name will consist of the abbreviated rank, first initial, and last name. The use of an individual's middle initial is optional. For extremely long names, it is permissible to use smaller letters to accommodate the entire name. (T-2)

4.5.1.6.13. Nicknames are not authorized. Size and font are at the MXG/CC's discretion; size not to exceed 2 ½ inches; standardized within the wing. Unit mascot graphics (i.e., razorback and eagle head [outlines or silhouettes]) may be used as the forward edge of the placard or crew chief block. For standardization purposes, either all or none of the heritage aircraft will bear the graphic. (T-2)

4.6. Approval Authority for Heritage Aircraft Markings.

4.6.1. Final approval for all temporary or permanent heritage markings in paragraph 4.5 will come from the Wing Commander. All levels of supervision have the responsibility to review the markings for tastefulness, appropriateness, and adherence to copyright laws.

4.7. Competition Aircraft.

4.7.1. Units participating in competitions will follow the guidelines established in competition rules for aircraft appearance. Competitions should be considered "come as you are" and no waivers will be granted. "Come as you are" is defined as no special effort, painting, or additional markings applied to enhance or improve the overall appearance of the aircraft. This includes polishing of metal surfaces, using commander type markings, etc. (T-2)

4.8. Aircraft Transfer.

4.8.1. The following markings must be removed prior to formal transfer of aircraft to other units or MAJCOMs (aircraft retiring to Aerospace Maintenance and Regeneration Group need not have any markings removed). Deviations from transfer requirements are authorized provided the gaining and losing units reach a mutual agreement. (T-3)

4.8.1.1. Organizational insignias.

4.8.1.2. Unit identifier.

4.8.1.3. Tail stripe.

4.8.1.4. Aircrew and crew chief names.

4.8.1.5. Unit unique markings.

4.8.1.6. Nose art.

4.9. Waivers.

4.9.1. Waiver requests must be approved by WG/CC, local Public Affairs, and Wing Historian prior to submission. E-mail the request using the AF Form 1768, Staff Summary Sheet, justification, design, photos, and tail number to NGB.A4.A4MM-Policy@us.af.mil for first coordination with the Command Fabrication Superintendent/Corrosion Manager. Once approved, the WG/CC will route the official request to NGB-CS through ETMS2, including all documents for final approval from NGB/A4. Waivers that are in violation of aircraft technical data will not be accepted. Waiver requests must include the following:

4.9.1.1. Clear statement of present procedure/marking.

4.9.1.2. Clear statement of proposed change.

4.9.1.3. Justification to include historical significance, if applicable.

4.9.1.4. Digital color photographs, one of present marking and one of requested change. The use of a slide presentation format is allowed.

4.9.1.5. Deviations from standard markings are authorized for AFSOC lead Wings. These Wings are authorized deviation for rapid deployment of assets.

Chapter 5
LOCAL MANUFACTURE PROGRAM

5.1. Local Manufacture Coordination.

5.1.1. To process local manufacture and/or modification request (Tools, Equipment, Aircraft Parts), the requesting squadron must complete all information on a locally generated Local Manufacture Coordination Sheet, or digital equivalent such as local SharePoint, including all applicable data (drawings, samples, TO references, engineering disposition, etc.), and route to the applicable fabrication section for review.

5.1.2. The Local Manufacture Coordination Sheet or digital equivalent such as local SharePoint will be utilized to facilitate coordination for all local manufacturing requests.

5.1.3. Local manufacture requests with a first and second position of the Source, Maintenance, and Recoverability (SMR) code identified as MO (an item to be manufactured or fabricated at the organizational maintenance level) or MF (An item to be manufactured or fabricated at the intermediate maintenance level) are, by technical order, authorized for local manufacture. SMR code applications and their definitions are outlined in TO 00-25-195, Source Maintenance, and Recoverability Coding of Air Force Weapons, Systems, and Equipment.

5.1.4. Parts SMR coded as procurable must be ordered through LRS/Materiel Management. However, should zero assets exist, and the item is deemed mission essential by MXG/CC or designated authority, and the fabricating section has the capability, the part may be manufactured. In that case, the requestor will contact the applicable engineering authority to obtain written authorization (e.g., memo for record, 202 or 107 approvals) to manufacture: Item Manager, System Program Office engineering.

5.2. Fabrication Flight Superintendent Responsibilities.

5.2.1. Must be designated approval authority per MXG/CC (Production Superintendent for off-shifts) to authorize manufacture and/or modification requests for aircraft, tools, and equipment items.

5.2.2. Act as final authority for manufacturing and/or modification requests supporting items that do not interfere with aircraft or equipment and are needed for internal maintenance.

5.3. Quality Assurance (QA) Responsibilities.

5.3.1. QA will be the final authority to authorize the use of manufactured and/or modified tools and equipment that interface with aircraft or equipment not identified or required by technical data. Coordination with QA must include a description of the item, pictures, drawings, samples (as needed), and/or engineering approval (if required).

5.3.2. Copies of all approved locally manufactured and/or modified tools and equipment requests will be maintained digitally by the QA office.

5.3.3. In the event a locally manufactured tool is designed to perform the same functions as a tool specified within a TO and the using shop determines that the locally manufactured tool is superior in function or use, QA shall submit the local tool to the Weapon System Program Office (SPO) for approval and inclusion within the TO. QA shall periodically follow up on the request until the request is either approved or rejected.

5.3.3.1. QA will keep the requestor informed on the status of the request and all communications concerning the request.

5.4. Requestor's Responsibilities.

5.4.1. Determine the local manufacture coordination and authorization process.

5.4.2. Order all aircraft parts/nationally stocked (NSN) items through LRS/material management using AF Form 2005, Issue/Turn-In Request.

5.4.2.1. Complete and sign the DD Form 1348M, DoD Single Line-Item Requisition System Document (Mechanical), generated by LRS/Material Management.

5.4.2.2. Generate an automated AFTO Form 350, Reparable Item Processing Tag or complete and provide a manual AFTO Form 350 tag.

5.4.2.3. Complete locally generated, Local Manufacture Coordination Sheet.

5.4.2.4. Obtain and provide all applicable data (drawings, samples, TO references, engineering dispositions, AF Form 2005, DD Form 1348-6, DoD Single Line-Item Requisition System Document, 350 tag, CAD model, etc.) and route to the applicable fabrication section for review.

5.4.2.5. Procure and deliver any/all required materials to the fabrication section to complete the request.

5.4.2.6. Process request for QA approval (as required) IAW paragraph 5.3.

5.4.2.7. The fabrication section will:

5.4.2.7.1. Evaluate all applicable data (drawings, technical data, samples) to verify manufacturing capability.

5.4.2.7.2. Provide a list of required materials on a locally generated Local Manufacturing Coordination Sheet or digital equivalent. The list will include any/all materials (tools, cutters, bits, consumables and/or expendables). Every effort will be made to utilize on-hand materials. Ordering of materials may be used to replenish stock. Assist with and provide procurement sources and costs for materials not available for acquisition through LRS.

5.4.2.7.3. Process the Local Manufacture Coordination Sheet or digital equivalent to the designated approval authority IAW paragraph 5.2.

5.4.2.7.4. Establish an estimated completion date according to assigned priorities. The completion date will be based on manufacturing time, availability of materials and/or estimated delivery of materials. The MXS Production Superintendent will establish the manufacturing priority for aircraft parts and tools.

5.4.2.7.5. Manufacturing and/or modification requests may be started immediately if all required materials are available. However, if the requestor must procure additional materials, the request will not be assigned or begin until all materials have been procured and delivered.

5.4.2.7.6. Track and coordinate actions required by additional work centers. Any additional maintenance action(s) conducted by additional work centers will be identified and documented on the locally generated Local Manufacture Coordination Sheet or digital equivalent. Throughout the process, the parts will be accompanied by all applicable data (drawings, samples, TO references, engineering dispositions, AF Form 2005, DD Form 1348-6, 350 tags, etc.)

5.4.2.7.7. Upon completion of the request, the primary work center assigned will ensure all maintenance actions for all work centers have been documented in the appropriate manner.

5.4.2.7.8. Notify the requestor the requested item is complete and ready for pick-up.

Chapter 6

AIRCRAFT BATTLE DAMAGE REPAIR (ABDR)

6.1. Aircraft Battle Damage Repair (ABDR).

6.1.1. ABDR is an effective force multiplier contributing to wartime sortie production by assessing and repairing battle-damaged aircraft rapidly to support flying operations. ABDR repairs will be accomplished during contingency or wartime only. However, weapons system program managers may approve ABDR repairs during peacetime on a case-by-case basis using trained ABDR Technicians.

6.2. Responsibilities.

6.2.1. The Directorate of Logistics (AF/A4L) provides overall policy and guidance for the USAF ABDR Program.

6.2.2. The Air Force Sustainment Center: Logistics, Directorate, Maintenance and Repair Requirements Branch (AFSC/LGPM) is the USAF ABDR Program Office and assumes management responsibility for USAF ABDR Programs. See DAFI 21-101_AFMCSUP, Aircraft and Equipment Maintenance Management, for guidance of Air Force Materiel Command (AFMC) responsibilities of the ABDR Program.

6.2.3. MAJCOM/A4M Responsibilities. Establish a command focal point to work ABDR issues with AFMC. The Command Fabrication Superintendent or designee will preside as the MAJCOM focal point for the ABDR program and perform the following duties:

6.2.3.1. Develop Command ABDR Concept of Operations (CONOPS) in coordination with AFMC and ABDR Technical Support Office (ABDR-TSO) to ensure the CONOPS addresses unit planning and readiness regarding the repair of battle/crash-damaged aircraft during combat operations.

6.2.3.2. Incorporate ABDR into Command war planning documentation.

6.2.3.3. Coordinate with AFMC for Unit Type Code (UTC) taskings in support of Operational Planning (OPLAN).

6.2.3.4. Address ABDR in mission need statements for new weapon systems that support or engage in combat operations.

6.2.3.5. Ensure ABDR bed down and integration requirements are outlined in the applicable Base Support Plan (BSP) IAW DAFI 10-404, Base Support and Expeditionary (BAS&E) Site Planning.

6.2.3.6. Provide unit-level weapon-system-specific tools (other than common hand tools) and equipment needed to repair battle/crash-damaged aircraft.

6.2.3.7. Provide technical support to the ABDR Technical Support Office for live fire or similar testing.

6.3. Maintenance Group Commander (MXG/CC) Responsibilities.

6.3.1. Ensure a minimum of 25% of 7 skill level or above maintenance personnel is trained and qualified to perform the role of Aircraft Battle Damage Evaluator using a mixture of personnel from 2A AFSCs. (T-3)

6.4. Unit Responsibilities.

6.4.1. Utilize trained Aircraft Battle Damage Evaluators to evaluate aircraft battle damage and mishap damage sustained during combat or contingency operations. (T-3)

6.4.2. Ensure shelf-life items listed in TO 1-1H-39, Aircraft Battle Damage Repair Manual – General, and weapon system-specific –39 TOs are maintained at required levels to support ABDR requirements. (T-3)

6.4.3. Ensure aircraft battle damage is documented on an AFTO Form 97, Aerospace Vehicle Battle Damage Incident Debrief/Assessment/Repair Record or AFTO Form 97B, Aircraft Battle Damage Evaluator Checklist as required IAW TO 1-1H-39. (T-2) Completed forms will be forwarded to the Aircraft Battle Damage Repair Program Office. (T-2) CLASSIFIED messages must be sent to SIPR: usafsa.wright-patt.afsc-lg.mbx.afsc-lgpm-abdrtso@mail.smil.mil and UNCLASSIFIED messages must be sent to NIPR: afsc.lgpm.abdrtso@us.af.mil for filing in the historical archives. (T-2)

6.5. Aircraft Battle Damage Evaluator Qualification and Training.

6.5.1. Aircraft Battle Damage Evaluator training provides MXG/CC's with ABDR knowledgeable forces and prepares units to execute Air Tasking Orders in a denied airspace with potential heavy losses. Aircraft Battle Damage Evaluator roles and responsibilities are outlined in TO 1-1H-39.

6.5.2. Must possess a minimum 7 skill level in an approved aircraft maintenance AFSC.

6.5.3. Complete Aircraft Battle Damage Evaluator training course every 24 months. (T-2)

6.6. Aircraft Battle Damage Evaluator Responsibilities.

6.6.1. Conduct Unexploded Explosive Ordinance (UXO) sweeps of aircraft and/or support equipment.

6.6.2. Ensure aircraft is safe for maintenance.

6.6.3. Debrief aircrew.

6.6.4. Evaluate sustained damage.

6.6.5. Coordinate repair effort with local supervision and ABDR Program Office.

6.6.6. Ensure completion of AFTO Form 97B, Aircraft Battle Damage Evaluator Checklist.

6.6.7. Locate and assist movement of ABDR War Reserve Material (WRM) assets.

6.7. Aircraft Battle Damage Evaluator Trainer Responsibilities.

6.7.1. Aircraft Battle Damage Evaluator formal training will be IAW the approved ABDR course control documents. (T-2) Ensure ABDR formal training complies with ABDR-TSO approved curriculum for field-administered training.

6.7.2. Field level Maintenance Training Flights (MTFs) will maintain instructors by using train-the-trainer. In the event an MTF has no trained instructors, contact the ABDR Program Office. (T-3)

6.7.3. Aircraft Battle Damage Evaluator instructor training material can be requested at afsc.lgpm.abdrtso@us.af.mil.

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

40 CFR, Part 61, *National Emission Standards for Hazardous Air Pollutants*

40 CFR, Part 63, *National Emission Standards for Hazardous Air Pollutants for Source Categories*

AFPD 21-1, *Maintenance of Military Materiel*, 01 August 2018

AFI 10-404, *Base Support and Expeditionary (BAS&E) Site Planning*, 24 July 2019

DAFI 21-101, *Aircraft and Equipment Maintenance Management*, 16 January 2020

DAFI 21-101_AFMCSUP, *Aircraft and Equipment Maintenance Management,* 8 November 2022

AFI 21-131, *Joint Oil Analysis Program*, 26 March 2014

AFI 33-322, *Records Management and Information Governance Program*, 23 March 2020

AFI 63-101/20-101, *Integrated Life Cycle Management*, 30 June 2020

DAFI 63-140, *Aircraft Structural Integrity Program and Air and Space Equipment Structural Management*, 6 August 2020

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AFMAN 48-148, *Ionizing Radiation Protection*, 20 July 2020

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DAFMAN 48-125, *Personnel Ionizing Radiation Dosimetry,* 27 October 2020

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DAFMAN 91-203, *Air Force Occupational Safety, Fire, and Health Standards*, 25 March 2022

TO 00-20-1, *Aerospace Equipment Maintenance Inspection, Documentation, Policies, and Procedures*, 26 September 2022

TO 00-25-107, *Maintenance Assistance,* 15 August 2022

TO 00-25-172, *Ground Servicing of Aircraft and Static Grounding/Bonding*, 22 December 2023

TO 00-25-195, *Source Maintenance, and Recoverability Coding of Air Force Weapons, Systems, and Equipment,* 1 September 2023

TO 00-25-252, *Intermediate Maintenance and Depot Level Maintenance Instructions - Aeronautical Equipment Welding*, 29 September 2023

TO 1-1A-1, *Engineering Handbook Series for Aircraft Repair, General Manual for Structural Repair,* 15 January 2016

TO 1-1A-9, *Engineering Series for Aircraft Repair Aerospace Metals - General Data and Usage Factors,* 7 December 2023

TO 1-1-689-3, *Cleaning and Corrosion Control, Volume III Avionics and Electronics*, 15 January 2021

TO 1-1-691, *Aircraft Weapons Systems - Cleaning and Corrosion Control and Prevention, Aerospace and Non-Aerospace Equipment*, 22 July 2022

TO 1-1-8, *Application and Removal of Organic Coatings, Aerospace and Non-Aerospace Equipment*, 24 September 2023

TO 1-1H-39, *Aircraft Battle Damage Repair Manual – General,* 15 May 2023

TO 1C-130A-6, *Aircraft Scheduled Inspection and Maintenance Requirements,* 15 December 2023

TO 1C-130J-6, *Airplane Scheduled Inspection,* 12 December 2023

TO 1C-135-3-8, *Location and Application - Exterior Stencils*, 15 May 2023

TO 32-1-101, *Use and Care of Hand Tools and Measuring Tools*, 19 August 2023

TO 33B-1-1, *Non-Destructive Inspection Methods, Basic Theory*, 01 July 2022

TO 33B-1-2, *Non-Destructive Inspection - General Procedures and Process Controls*, 15 December 2023

TO 34-1-3, *Inspection and Maintenance - Machinery and Shop Equipment*, 23 August 2019

TO 34A-1-1, *Additive Manufacturing Qualification of Technicians, Machines and Facilities,* 20 December 2020

TO 34A-3-1, *Polymers Additive Manufacturing, General Procedures and Process Controls*, 30 December 2020

TO 35-1-3, *Corrosion Prevention and Control, Cleaning, Painting, and Marking of USAF Support Equipment (SE)*, 26 January 2022

Prescribed Forms

AMC Form 1017*, Aircraft Wash Supervisor and Employee's Certification*

Adopted Forms

DD Form 1348-6, *DoD Single Line-Item Requisition System Document*

DD Form 1348M, *DoD Single Line-Item Requisition System Document (Mechanical)*

DD Form 2757, *Welding Examination Record*

DAF Form 679, *Department of the Air Force Publication Compliance Item Waiver Request/Approval*

DAF Form 847, *Recommendation for Change of Publication*

AF Form 1768, *Staff Summary Sheet*

AF Form 2005, *Issue/Turn-in Request*

AFTO Form 1, *Welding Proficiency Log*

AFTO Form 95, *Significant Historical Data*

AFTO Form 97, *Aerospace Vehicle Battle Damage Incident Debrief/Assessment/Repair Record*

AFTO Form 97B, *Aircraft Battle Damage Evaluator Checklist*

AFTO Form 350, *Reparable Item Processing Tag*

AFTO Form 781A, *Maintenance Discrepancy and Work Document*

Abbreviations and Acronyms

ABDR—Aircraft Battle Damage Repair

ABDR-TSO—Aircraft Battle Damage Repair-Technical Support Office

ADCC—Assistant Dedicated Crew Chief

AFCPCO—Air Force Corrosion Prevention and Control Office

AFI—Air Force Instruction

AFLCMC—Air Force Life Cycle Management Center

AFMAN—Air Force Manual

AFMC—Air Force Materiel Command

AFNDIO—Air Force Nondestructive Inspection Office

AFOUA—Air Force Outstanding Unit Award

AFRC—Air Force Reserve Command

AFSC—Air Force Specialty Code

AGE—Aerospace Ground Equipment

AIRCAT—Automated Inspection, Repair, Corrosion, and Aircraft Tracking

ALC—Air Logistics Center or Air Logistics Complex

AM—Additive Manufacturing

AMPO—Additive Manufacturing Program Office

AMC—Air Mobility Command

AMT—Aircraft Metals Technology

ANG—Air National Guard

ARC—Air Reserve Component

ASIP—Aircraft Structural Integrity Program

ASM—Aircraft Structural Maintenance

AvSE—Aviation Support Equipment

BSP—Base Support Plan

CBT—Computer Based Training

CCPE—Corrosion Control Prevention Executive

CFR—Code of Federal Regulations

CONOPS—Concept of Operations

COR—Contracting Officer Representative

CPA—Corporate Process Activities

CPAB—Corrosion Prevention Advisory Board

CPC—Corrosion Preventative Compound

CS—Cold Spray

DAFI—Department of the Air Force Instruction

DAFMAN—Department of the Air Force Manual

DCC—Dedicated Crew Chief

DLA—Defense Logistics Agency

DPAS—Defense Property Accountability System

EID—Equipment Identification Designator

ETMS2—Enterprise Task Management Software Solution

FMxC2—Field Maintenance Command and Control

FRP—Fuselage Reference Plane

GMAW—Gas Metal Arc Welding

GTAW—Gas Tungsten Arc Welding

HSS—High-Speed Steel

IAW—In Accordance With

ICARR-3D—Inspection, Crack/Corrosion and Repair Reporting

IMI—Interactive Multimedia Instruction

IPB—Illustrated Parts Breakdown

IPT—Integrated Process Teams

ISO—Isochronal Inspection

JEDMICS—Joint Engineering Data Management Information and Control System

M&U—Maintenance and Utilization

MAJCOM—Major Command

MDS—Mission Design Series

MFTO—Metals Fabrication and Technology Office

MIS—Maintenance Information System

MTF—Maintenance Training Flight

MUA—Meritorious Unit Award

NAS—National Aerospace Standards

NAS 410—National Aerospace Standard Certification & Qualification of Nondestructive Test Personnel

NDI—Nondestructive Inspection

NGB—National Guard Bureau

NLT—No Later Than

NSN—National Stock Number

OAP—Oil Analysis Program

OPLAN—Operational Planning

OPR—Office of Primary Responsibility

OWS—Outer Wing Station

PCAMS—Process Control Automated Management System

PCR—Publication Change Request

PCS—Permanent Change of Station

PE—Personal Evaluation

PIT—Product Improvement Team

POC—Point of Contact

POM—Program Objective Memoranda

PPE—Personal Protective Equipment

PS&D—Plans, Scheduling, and Documentation

QA—Quality Assurance

QPD—Qualified Product Database

QPL—Qualified Products Listings

QVI—Quality Verification Inspections

RIL—Routine Inspection Listing

RSO—Rapid Sustainment Office

RVSM—Reduced Vertical Separation Minimum

SE—Support Equipment

SEMIS—Support Equipment Maintenance Information System

SI—Special Inspections

SII—Special Interest Items

SMAW—Shielded Metal Arc Welding

SME—Subject Matter Expert

SMR—Source Maintenance, and Recoverability

SNCO—Senior Noncommissioned Officer

SPO—System Program Office

STRT—Specialty Training Requirements Team

TAR—Technical Assistance Request

TDP—Technical Data Package

TDY—Temporary Duty

TO—Technical Order

U&TW—Utilization and Training Workshop

UTC—Unit Type Code

UXO—Unexploded Explosive Ordinance

VSS—Vertical Stab Station

W&B—Weight and Balance

WRM—War Reserve Material

WSM—Weapon System Manager

Office Symbols

AF/A4L—The Directorate of Logistics

AFSC/LGPM—Air Force Sustainment Center; Logistics, Directorate, Maintenance and Repair Requirements Branch

AMC/A4—Air Mobility Command; Directorate of Logistics, Engineering, and Force Protection

AMC-CS—Air Mobility Command; Chief of Staff

AMC/A4M—Air Mobility Command; Maintenance Division

MXS/CC—Maintenance Squadron Commander

MXG/CC—Maintenance Group Commander

WG/CC—Wing Commander

Attachment 2
MARKING LOCATION BY AIRFRAME

Table A2.1. Marking Specifications.

Aircraft Table	
A-10	A2.2.
HC/MC/MP/-130	A2.3
C/EC/-130 E-J	A2.4.
E-8(B-707-400)	A2.5.
F-15	A2.6.
F-16	A2.7.
F-22 & B-2	A2.8.
F-35	A2.9.
HH-60	A2.10.
MQ-1	A2.11.
MQ-9	A2.12.
C-17	A2.13.
KC-135	A2.14.
C-21	A2.15.
C-27J	A2.16.
NOTES: Not all inclusive, refer to specific weapon system T.O. or drawings for further guidance.	
1. The word “Optional” means a local Wing Commander option. Local markings will not be placed on the aircraft during PDM or contractor scuff sand and repaints. Units must apply at home station.	
2. The use of computer stencil makers, and vinyl material at local units and Depot/Contractors is authorized IAW T.O.1-1-8. Variations in size, width, length, and spacing of letters/numbers may be different due to the various computer programs available.	
3. Associate Wings may apply their Command or Squadron emblems on ANG owned aircraft. ANG emblems will always be first. Location and size of these markings will be determined by the owning unit.	

Table A2.2. A-10C

COMMAND MINUTEMAN INSIGNIA: 18 inches black (flat black) OPTIONAL.
TBD by Wing Commander, (tail or fuselage).
Vertical: 10 inches below bottom edge of rudder cap.
Horizontal: Centered between leading edge (LE) and first row of rivets aft of LE.
ORGANIZATIONAL INSIGNIA: 18 inches black (flat decal or colors) Optional.
LT side: Above panel F-18 and aft of panel F-44.
RT side: Above panel F-79 and aft of panel F-105.
UNIT IDENTIFIER: 12 inches (flat black or contrasting gray) Optional.
Vertical: Lower edge 3 inches above tail numbers.
Horizontal: Centered on vertical stabilizer.
PILOT AND CREW CHIEF NAMES: (flat black) Optional.
Pilot on LT side under windscreen beginning at FS 188.92.
Crew chief just under pilot's name. Assistant crew chief name just under crew chief name.

Table A2.3. HC/MC/MP-130.

HC Aircraft assigned to the 106, 129 and 176 RQW are now ACC gained, but painted in the AFSOC paint scheme, using AFSOC Drawing #93104893.
Paint and markings will be two tones: #36118/dark gray and #36293/light gray.
The US Flag are not authorized on MC and MP aircraft. ACC must approve their application on HC aircraft.
All other marking will be Wing Commander Options. Maintenance and Operations need to communicate before these options are accomplished to ensure mission integrity is not compromised.
ANG Tail Marking: 6 inches (Flat Black) Optional Centered on both sides of vertical stabilizer above the tail number.
COMMAND MINUTEMAN INSIGNIA: 30 inches, (contrasting gray) Optional. Location TBD (tail or fuselage).
ORGANIZATIONAL INSIGNIA: 30 inches (contrasting gray) Optional. Insignia will be placed on the fuselage, the bottom of the insignia will center on WL 190.0. The insignia will be centered on FS 270.0.

STATE NAME: 10 inch (contrasting gray) Optional. Centered on FS 551.00, 10 inches from main landing gear door hinge point.
PILOT/CREWCHIEF/ASSISTANT Names: (contrasting gray) Optional. Size and location determined by the unit. Placing names inside the aircraft will avoid removal and replacement prior to and after deployments.
AIR FORCE OUTSTANDING AWARD: 3 inches by 12 inches (flat decal, contrasting gray). Centered, 3 inches above crew entry door Optional.
ENGINE INLET SCOOPS and TI APU Panels: Optional. These areas may be stripped if paint erosion or heat damage is occurring. Submit TAR to C-130 Engineering for disposition.
BLACK EXHAUST TRACKS: Optional. Black exhaust tracks may be added to the bottom of the wing and flap areas only. This was approved by C-130 Engineering and AFSOC HQ, as a field level option only. Black exhaust track will not exceed 96” wide.

Table A2.4. C/EC/-130 E-J

Equipment Excellence aircraft assigned to the 176 AW are authorized to use tail marking configuration AK in flat black or contrasting gray. (See Attachment 7).
Equipment Excellence EC-130J aircraft assigned to the 193 SOW will be painted IAW AF Drawings # 200917976 and # 200917977).
Equipment Excellence C-130J aircraft will be painted IAW Drawings # 201122423 and # 201122424 or contact C-130 SPO if additional info is required.
High Gloss LC-130 aircraft assigned to the 109th AW will be painted IAW AF Drawing 9144927.
Wheeled aircraft will be painted Equipment Excellence, IAW AF Drawings 9144700, 9276080, 9276081, 9276082.
UNITED STATES FLAG: 24 inches X 48 inches (matte finish). Both sides vertical stabilizer, bottom of flag located 154 inches above horizontal stabilizer with bottom of flag centered horizontally on vertical stabilizer. (Except EC).
ANG, TAIL MARKING: 12 inches (flat black). Both sides of vertical stabilizer, top of letters located 10 inches below and centered under flag. (Except EC).
VERTICAL STAB TIP STRIPE: Top 30 inches of vertical stabilizer to run horizontally not to exceed 15inches. Solid color with no other marking allowed. Optional (Except EC).
TAIL BAND STRIPES: (flat black) 2-inch upper strip located 10 inches below bottom of ANG; 2 inches lower stripe located 12 inches below bottom of upper stripe. Top horizontal

stripe will wrap around leading edge and run to the trailing edge of rudder, not to extend onto the rudder trim tabs; bottom stripe will wrap around the leading edge and run to the trailing edge of the rudder. (See Attachment 7) (Except EC)
RADIO CALL NUMBERS: 12 inches (flat black). Both sides of vertical stabilizer, top of numbers located 10 inches below bottom of lower tail band stripe, centered under flag. (See Attachment 7) (Except EC)
NOSE CALL NUMBERS: 4 digit, 6 inches (flat black) Optional. Location starting at FS 139 (measurement is for H/J model aircraft and may vary on older E aircraft) and runs aft. The bottom of numbers is horizontal with the bottom of the lower pilot window WL 200.00 to 198.00.
UNIT CALL NUMBERS/LETTERS: 6 inches (flat black) Optional. Top of number/letter is located 6 inches from the bottom of the nose call number, starting at FS 139.
AIR FORCE OUTSTANDING AWARD: 3 inches by 12 inches (flat decal, contrasting grays or flat black), centered 3 inches above crew entry door. Optional.
PILOT/CREW CHIEF/ASSISTANT NAMES: (flat black) Optional. Size and location determined by the unit. (Except EC). Placing names inside the aircraft will avoid removal and replacement prior to or after deployments.
COMMAND MINUTEMAN INSIGNIA: 30 inches, (flat black or gray decal) Optional. Both side of vertical stabilizer centered with the flag. The top of the emblem is 10 inches from the bottom of the call number.
STATE NAME: 10 inches (flat black), centered on FS 551.00 and 10 inches from main landing gear door hinge point. (Attachment 7)
BLACK EXHAUST TRACKS: Optional. C-130 engineering has approved, at field level, the widening of exhaust tracks from 60 inches to 96 inches, on the bottom of the wing and flap area. Application is also extended to the top of the flaps, if desired. SPD approval is on file at NGB/A4MM.
NOTE: IAW 1C130A-23CL, use of CBA6 Soil Barrier in the exhaust area is a Command Option. Units are no longer mandated to use this material.

Table A2.5. E-8 (B-707-400).

American Flag: 31.5 inches by 60 inches. Applied to both sides of the vertical stabilizer. The top of the flag is located at Fin Station 210.15 with the forward top corner of each flag on the vertical leading-edge seam.

COMMAND & ORGANIZATIONAL INSIGNIA: 18 inches (gloss, multi-color), Optional. Insignias will be placed on the left side of the aircraft, aft of the fwd crew door in the following order: ANG shield, 116 ACW shield, and ACC shield.
Vertical: 1-inch FWD of center of pilot/copilot #2 window.
Horizontal: 11 inches below lowest part of sill.
UNIT IDENTIFIER: 36 inches (gloss black) Optional. The GA will be placed on both sides of the vertical tail, 24 inches below the American flag.
LT Side: Designator will begin 20 inches in from LE.
RT Side: Designator will end 20 inches in from LE.
Radio Call Numbers: 12 inches high (gloss black). Bottom of number, 33 inches above WL 303.30, top corner of the first number (left hand side) or last number (right hand side) at the leading-edge seam.
PILOT AND CREW CHIEF/ASSISTANT NAMES: 3 inches in height maximum (gloss black) Optional.
Pilot and co-pilot will be placed on the left side of the aircraft nose under the windows: Crew Chief and Asst Crew Chief will be placed on the right side of the aircraft nose under the windows. MCC and AOT crew names will be posted on the inside of the aircraft. Location TBD by unit. Crew chief/assistant names: 18-inch by 36-inch block, 10 inches aft of crew entry door, top of block flush with top of door.
NOSE NUMBER: 6 inches (gloss black). Optional. Last four digits of tail number, on left and right nose gears doors, 2 inches from bottom of the door and centered between fwd and aft edges.
AIR FORCE OUTSTANDING AWARD: 3 by 12 inches (gloss, multi-color decal), Optional. Located on the left side of the aircraft, 4 inches above and centered on the 116 ACW shield.
TAIL STRIPE: 15 inches in width. (Gloss-multi-color), Optional.
The background is black, (color #17038), 11 inches wide, with two red (color # 11136), 2-inch stripes located on the top and bottom of the black background. The GEORGIA will be in Gold Metallic finish (color #17043) and the letters are BOOKMAN BOLD ACCT. A.K. Rev C, with a 15-degree slant (facing aft for both left and right side), 9 inches high with a one-inch gap between top and bottom red stripes.
LOCATION: 20 inches below the top of the vertical stabilizer and in a leveled position.

Table A2.6. F-15

COMMAND MINUTEMAN INSIGNIA: 18 inches (flat black) Optional. Location TBD (tails or fuselage).
ORGANIZATIONAL INSIGNIA: 18 inches (flat decal or flat colors) Optional. Vertical: Bottom of insignia on WL 100.0. Horizontal: Forward edge of insignia on FS 458.0.
UNIT DESIGNATOR: 24 inches (flat black) Optional. Vertical: Top of letters even with top of rudder. Horizontal: Leading edge of first letter on FS 760.0.
PILOT and CREW CHIEFS' NAMES: Size TBD by unit. Optional. Pilot name centered below left windscreen frame and crew chief, and assistant crew chief names centered below right wind screen frame.
AIR FORCE OUTSTANDING AWARD: 3 inches by 12 inches (flat decal, contrasting grays or flat black). Location TBD by each unit. Optional
NOSE NUMBER: 4 inches (flat black). Optional. Last three or four digits of tail number vertically on left and right side of nose gear door.
RADIO CALL NUMBER: 15 inches (flat black), location IAW 1F-15A-23, Fig 10-l.

Table A2.7. F-16.

COMMAND MINUTEMAN INSIGNIA: 18 inches (flat black). Optional. Location TBD by Wing Commander, (tail or fuselage).
ORGANIZATIONAL INSIGNIA: 10 inches (subdued decal or flat colors). Optional. Vertical: Top of insignia 11 inches below fuselage/intake splitter vane. Horizontal: Leading edge 52 inches aft of intake duct lip.
UNIT DESIGNATOR: 18 inches (flat black). Optional. Vertical: Bottom of numbers at WL 158.0. Horizontal: Leading edge of first letter on FS 482.07.
AIR FORCE OUTSTANDING AWARD: 3 inches by 12 inches (flat decal, contrasting grays or flat black), Location TBD by each unit. Optional.
PILOT and CREW CHIEFS' NAMES: Size TBD by unit. Optional. Pilot name on left canopy rail, and crew chief and assistant crew chief names on right canopy rail.
NOSE NUMBER: 4 inches (flat black); last three or four digits of tail number on both sides of nose gear door. Optional.

RADIO CALL NUMBER: 12 inches (contrasting gray) NOTE: Field units may change call numbers to flat black at home station. Aircraft returning from depot will be painted in contrasting gray IAW 1F-16C-2-00GV-00-1.

Table A2.8. F-22 and B2.

External Marking will be applied IAW ACCI 21 105.
NOTE: Hawaii F 22 aircraft are authorized to use the PACAF Tail Marking configuration "HH" on the aircraft. For all other external aircraft markings follow guidance: T.O. 1F22A 200GV001, General Vehicle Description – F22A Aircraft.

Table A2.9. F-35.

COMMAND INSIGNIA: 12 inches (contrasting shade of gray). Bottom of the insignia will be centered 47 inches above the bottom edge of the blackboard and centered FWD to aft in the vertical stabilizer blackboard area.
WING INSIGNIA: 12 inches (contrasting shade of gray). Insignia will be centered within the blackboard area on the RT side inlet below the chine and formation light.
SQUADRON INSIGNIA: 12 inches (contrasting shade of gray). Insignia will be centered within the blackboard area on the LT side inlet below the chine and formation light.
UNIT IDENTIFIER: 12 inches (contrasting shade of gray). Bottom of the designator will be centered 25 inches above the bottom edge of the vertical stabilizer blackboard.
TAIL STRIPE: Not authorized for F-35 aircraft.
RADIO CALL NUMBERS: 6 inches (contrasting shade of gray). Bottom of the tail number will be centered 2 inches above the bottom edge and 2.6 inches from the inner most trailing edge corner of the vertical blackboard area.
UNIQUE MARKINGS: Commander's Flagship: WG, OG, height is 6 inches (contrasting shade of gray) Bottom of the identifier will be 14 inches above the bottom edge of blackboard area, centered between FWD and aft blackboard edges.
PILOT AND CREW CHIEF NAMES: 2 inches Pilot: End of name will be located two inches FWD of IB aft apex BL 0 and 2 inches from BL 0 door edge (LT NLG door). Crew Chief: Beginning of name/rank will be located 2 inches FWD of IB aft apex (BL 0) and 2 inches from BL 0 edge (RT NLG door).
NOSE NUMBERS: 4 inches (contrasting shade of gray). Nose numbers will be located four inches from the inboard door FWD apex (BL 0) and two inches from BL 0 door edge.

NOTE: All lettering/numbering applied to F-35 blackboard areas will meet vertical block type/style font/lettering and Arabic numerals as specified in T.O. 1-1-8, Appendix B, Figure B-3 (Form of Letters and Numerals) or similar computer-generated font.

Table A2.10. HH-60.

Aircraft assigned to the 106, 129 and 176 are now ACC gained and will be painted with MilPRF-85285 Extended Life Coating, Color # 36118 and marked in Flat Black color # 37078.
COMMAND MINUTEMAN INSIGNIA: 10 inches (flat black). Optional. Location TBD by Wing Commander.
ORGANIZATIONAL INSIGNIA: 10 inches (subdued decal or flat colors). Optional. Wing: on right cargo door 8 inches below forward window, centered. Squadron: on left cargo door, 8 inches below forward window, centered.
PILOT/AIRCREW/CREW CHIEF NAMES: 3 inches maximum (flat black). Optional. Pilot: Right door, 2.5 inches below window, centered. Copilot: Left door, 2.5 inches below window, centered. Crew chief/assistant: Crew chief, right cargo door, 3.1 inches below and centered on forward window. Assistant: Left cargo door, 3.1 inches below and centered on forward window. Placing names inside the aircraft will avoid removal and replacement prior to or after deployments.
AIR FORCE OUTSTANDING AWARD: 3 inches by 12 inches (flat decal, contrasting gray or flat black), Location TBD buy each unit. Optional.

Table A2.11. MQ-1.

ANG Command Insignia: 10 inches (flat black vinyl).
Vertical: Top of insignia 4.5-inches below bottom-edge of antenna panel seam. Horizontal: Centered between leading edge of wing and aft edge of satcom radome.
Organizational Insignia: 10 inches (flat black vinyl). Optional.
Vertical: Top edge of insignia 4.5 inches below bottom edge of antenna panel seam. Horizontal: Centered between leading edge of wing and aft edge of satcom radome.
Unit Designator: 10 inches (flat black vinyl). Optional. Centered vertically and horizontally on tail outboard surfaces; leveled perpendicular to tail plane trailing edge.
Squadron Insignia/Patch: 10 inches (flat black vinyl). Optional.

Vertical: Bottom edge of insignia 4.5 inches above chine line left side of fwd fuselage.
Horizontal: Centered between leading edge of wing and aft edge of forward satcom radome. Commander
Flagship: Unit designator: 12 inches (flat black with white shadow to aft side).
Vertical: Top edge of marking 25 inches below top edge of stabilizer.
Horizontal: Centered between leading and trailing edge surfaces on upper vertical tail leveled parallel with tail stripe.
Air Force Chevron: 10 inches (flat black vinyl).
Fuselage placement: Located on left and right sides of aft fuselage.
Vertical: Approximately 6 inches above chine line. Horizontal: 10 inches from trailing edge.
Wing placement: Located on top of left wing and bottom of right wing.
Placement: Centered on chord line 8 feet from wing tip. Alignment: Parallel to the main wing spar line. Tail Stripe: 6 inches wide, located on outside of the left and right tail boards.
Vertical: Top edge of stripe 75 inches below top edge of installed tail board.
Alignment: Perpendicular to tail board trailing edge. Crew Names: Up to 1.75-inch block letters (flat black vinyl).
Pilot: Located on left side of forward fuselage
Vertical: 1 inch below front satcom radome.
Horizontal: Centered between aft and forward edges of front satcom radome.
Sensor Operator (SO): Located on left side of forward fuselage. **Vertical:** Up to 3.25 inches below front satcom radome (0.5 inch below pilot name). **Horizontal:** Left edge aligned with left edge of pilot name or centered under pilot name, which ever presents the most professional appearance based on individual name lengths. **Crew Chief:** Located on right side of forward fuselage. **Vertical:** 1 inch below front satcom radome. **Horizontal:** Centered between aft and forward edges of front satcom radome.
Assistant Crew Chief: Located on right side of forward fuselage. **Vertical:** Up to 3.25 inches below front satcom radome (0.5 inch below CC name).
Horizontal: Centered under CC name. Tail Number: Located on the left and right sides of the aft fuselage. 1.5-inch block letters AF over 1.5-inch numbers (year designator) followed by 4-

inch numbers (last three numbers in aircraft serial number) (Flat black vinyl). Propeller Tip Blade: 4-inch band around blade tip (flat yellow paint). *(Optional)
U.S. AIR FORCE or "State" AIR GUARD: 7 inches (flat black), location: both sides of fuselage located between Air Force Chevron and Organization Insignia on the left side and between the Air Force Chevron and the ANG Command Patch on the right side. Both will be centered from top to bottom on the fuselage.

Table A2.12. MQ-9.

ANG Command Insignia: 12 inches (flat black vinyl). Vertical: Top of insignia 12-inches below bottom-edge of tail stripe on upper tails. Horizontal: Centered between leading and trailing edges of upper tails with top edge perpendicular to trailing edge.
Organizational Insignia: 10 inches (flat black vinyl) Optional. Vertical: Bottom edge of insignia 4 inches above chine line right side of fwd fuselage. Horizontal: Centered between leading edge of wing and aft edge of satcom radome.
Unit Designator: 10-inch block letters (flat black vinyl) Optional. Vertical: Bottom edge of "NY" 61 inches from base of upper tails. Horizontal: Centered between leading and trailing edge surfaces on upper vertical tail perpendicular with satcom radome.
Squadron Insignia/Patch: 10 inches (flat black vinyl) Optional. Vertical: Bottom edge of insignia 4 inches above chine line left side of fwd fuselage. Horizontal: Centered between leading edge of wing and aft edge of satcom radome.
Commander Flagship: Unit designator: 10 inches (flat black with white shadow to aft side) Vertical: Bottom edge of ―NY‖ 61 inches from base of upper tail. Horizontal: Centered between leading and trailing edge surfaces on upper vertical tail leveled parallel with tail stripe. 174FW Shadowed to the AFT side, 44 inches from the base of upper tail.
Tail Number: 3.5-inch block letters AF over 3.5-inch numbers (year designator) followed by 8-inch numbers (last three numbers in aircraft serial number) (Flat black vinyl, shadowed white to the AFT)
Air Force Chevron: Fuselage placement: 10 inches (flat black vinyl) Located on left and right sides of aft fuselage. Vertical: 4 inches above chine line. Horizontal: Centered between wing trailing edge and upper tail leading edge. Wing placement: Located on top of left wing and bottom of right wing. Placement: Centered on chord line 12 feet from wing tip to outer edge of decal. Alignment: Parallel to the main wing spar line.
Tail Stripe: 7 inches wide located on left and right upper tails. Vertical: Top edge of stripe 4 inches below top edge of upper tails. Horizontal: Perpendicular to tail trailing edge.
Aircrew Names: 1.75-inch military block letters (flat black vinyl). Pilot: Located on left side of forward fuselage. Vertical: 2 inches below chine line. Horizontal: Centered on satcom

radome Sensor Operator (SO): Located on left side of forward fuselage. Vertical: 0.5 inches below pilot name. Horizontal: Centered below pilot name.
Ground Crew Names: 1.75-inch military block letters (flat black vinyl). Crew Chief (CC): Located on right side of forward fuselage. Vertical: 2 inches below chine line. Horizontal: Starting at 26 inches forward of aft edge of satcom radome justified forward. Assistant Crew Chief (ACC): Located on right side of forward fuselage. Vertical: 0.5 inches below CC name.
Tail Number: 3.5-inch block letters AF over 3.5-inch numbers (year designator) followed by 8-inch numbers (last three numbers in aircraft serial number) (Flat black vinyl). Horizontal: Located on the left and right sides of the lower tail leveled 4 inches below fuselage. Alignment: Centered between leading and trailing edge of rudder. Propeller Blade Tips: 4-inch band around blade tips (flat yellow paint).

Table A2.13. C-17.

UNITED STATES FLAG: 24 inches by 48 inches (Matte). Both sides vertical stabilizer. Bottom of flag located 42 inches above top edge of the of the upper tail band stripe, with the top forward corner of the flag located 1 inch from the VOR/LOC-2 antenna.
ANG, TAIL MARKING: 18 inches (flat black). Both sides of vertical stabilizer, bottom of letters located 12 inches above top edge of the top tail band stripe and centered on an (invisible) vertical line drawn parallel with vertical stabilizer trailing edge that intersects the center of the flag.
TAIL BAND STRIPES: (flat black) Two 2-inch stripes, top of upper stripe located at vertical stabilizer coordinate ZV 134. Top of lower stripe is located 18 inches below bottom of upper stripe. Stripes run horizontally from aft edge of leading-edge seam to trailing edge of rudder.
RADIO CALL NUMBERS: 12 inches (flat black). Both sides of vertical stabilizer, top of numbers located 12 inches below bottom of lower tail band stripe, centered on an invisible vertical line drawn parallel with the vertical stabilizer trailing edge, intersecting center of the flag.
LOCAL STATION NUMBERS: 18 inches (flat black). Optional. Last 4 digits of aircraft serial number, located on both sides of the fuselage, centered below the lower aft corner of the down view window, with the top of the numbers on fuselage coordinate Z-192.
UNIT IDENTIFIER: 10 inches (flat black). Optional. Both sides of fuselage, centered on the station number, top of numbers 6 inches below the bottom of the station number.
ASSOCIATE UNIT IDENTIFIER: 10 inches (flat black). Optional Both sides of the fuselage, centered on the unit identifier, top of numbers 6 inches below bottom of the unit designator.

AIR FORCE OUTSTANDING AWARD: 3 inches by 12 inches (flat decal, contrasting grays or flat black), centered 3 inches above crew entry door. Optional.
PILOT/CREW CHIEF/ASSISTANT NAMES: (flat black). Optional. Names can be located on either side of the fuselage. Size and location determined by the unit. Placing names inside the aircraft will avoid removal and replacement prior to or after deployments.
COMMAND MINUTEMAN INSIGNIA: 34 inches (flat black or gray decal). Optional. Both sides of fuselage, most forward edge of emblem located 3 inches aft of fuselage light ring, bottom tip of emblem almost touching longeron L-32.and even with top of crew entry door.
STATE NAME: Size TBD by unit (flat black), centered on both sides of main landing gear wheel well assembly.
NATIONAL STAR INSIGNIA OUTLINE: 30 inches (flat black); both sides of fuselage, centered on centerline of the aft fuselage formation light, with the insignia leading edge located 6 inches aft of the light. U.S. AIR FORCE MARKING: 24 inches (flat black); both sides of fuselage, located 12 inches aft of fuselage station 27.200 and 35.38 inches above longeron 1-25.

Table A2.14. KC-135.

Aircraft assigned to Hawaii and Alaska authorized tail marking configuration (HH or AK), in flat black, # 37038 on aircraft painted Equipment Excellence, color # 36173.
UNITED STATES FLAG: 21 inches by 40 inches (Matte); both sides vertical stabilizer, bottom of flag on WL 447, centered between stabilizer leading and trailing edges, not to include rudder.
ANG TAIL MARKING: 12 inches (flat black); both sides of vertical stabilizer, centered between stabilizer leading and trailing edges, not including rudder.
TAIL BAND STRIPES: (flat black). Optional. Two-inch upper strip grounded at WL 568.90, top of the lower 2-inch stripe located 12 inches below the bottom of the upper stripe. Unit tail band applied at home station; will be completely contained within the black line area prescribed above.
RADIO CALL NUMBERS: 12 inches (flat black); both sides of vertical tail, top of numbers located 12 inches below ANG tail marking, centered between stabilizer leading and trailing edges, not including rudder.
LOCAL STATION NUMBERS: 6 inches (flat black) Optional. Last 4 digits of aircraft serial number, both sides of fuselage nose section. Locate according to T.O. 1C-135-8.

UNIT IDENTIFIER: 6 inches (flat black) Optional; both sides of fuselage centered 6 inches under station number. ASSOCIATE
UNIT IDENTIFIER: 6 inches (flat black). Optional Both sides of the fuselage. Located 6 inches below bottom and centered on the unit designator. AIR FORCE OUTSTANDING AWARD: 3 inches by 12 inches (flat decal, contrasting grays or flat black) centered 3 inches above crew entry door. Optional.
PILOT/CREW CHIEF/ASSISTANT NAMES: (flat black) Optional. Names can be located on either side of the fuselage. Size and location determined by the unit. Placing names inside the aircraft will avoid removal and replacement prior to or after deployments.
COMMAND MINUTEMAN INSIGNIA: 34 inches (flat black or gray decal) Optional; both sides of fuselage, 16 inches aft of crew entry door, 6 inches below USAF markings.
BOOM ELEVATORS: 10 inches (Color # 36622). Highest numeric designator of station assigned centered on the underside of the left ruddevator and alpha designator (ANG) centered on underside of the right ruddevator. Top of both ruddervators marked in the obverse of the bottom.

Table A2.15. C-21

UNITED STATES FLAG: 5 inches (Gloss) both sides of vertical stabilizer 12 inches up from the top surface of the HF Antenna and 15.5 inches forward of the rudder attachment point.
TAIL BAND and STRIPES: 1 inch gloss gold stripes 8-inch band w/gloss multi-colors 6-inch gloss lettering. Both sides of the vertical and rudder. Lettering centrally located vertically and horizontally within the tail band parallel to the aircraft centerline. Tail band starts 2 inches below the US flag. Note: If tail band and stripes extend onto rudder area rudder must be re-weighed and balanced IAW TO or aircraft manufacture instructions.
ANG EMBLEM: 12 inches placed on the left side of the forward fuselage 5 inches below the corner of the windshield and centered between the edge on the entrance door and windshield.
RADIO CALL NUMBERS: 8 inches. Both sides of vertical stabilizer 9 inches below lower surface of the HF antenna and 15.5 inches forward of the rudder.
PILOT/CREW CHIEF NAMES: 1 inch gloss black. Pilots/crew chief names left nose landing gear door 2 inches aft of forward edge.

Table A2.16. C-27J.

UNITED STATES FLAG: 20 inches X 38 inches (Matte finish). Located on both sides vertical stabilizer and placed 10 inches above the top edge of the MAJCOM Designator.

Should be horizontally centered between the vertical stabilizer leading edge and the rudder hinge point.
ANG, TAIL MARKING: 12 inches (37038) applied halfway (centered) between the bottom and top edges of the vertical stabilizer. It should be horizontally centered between the vertical stabilizer leading edge and the rudder hinge.
TAIL BAND STRIPES: (37038) Two 2-inch stripes, 12 inches apart, running horizontally across the tail and the rudder. The top edge of the top stripe should be 10 inches below the bottom edge of the MAJCOM designator.
RADIO CALL NUMBERS: 10 inches (37038). Both sides of vertical stabilizer, top of numbers located 10 inches below bottom edge of the tail stripe. Numbers should be horizontally centered between the vertical stabilizer leading edge and the rudder hinge point.
NOSE CALL NUMBERS: 4 digits, 6 inches (37038) Optional. Location is below the pilot's window.
UNIT CALL NUMBERS/LETTERS: 6 inches (37038) Optional. Top of number/letter is located 6 inches from the bottom of the nose call number. Location is below the pilot's window.
AIR FORCE OUTSTANDING AWARD: 3 inches by 12 inches (flat decal, contrasting grays or flat black), centered 3 inches above crew entry door. Optional.
PILOT/CREW CHIEF/ASSISTANT NAMES: (37038) Optional. Size and location determined by the unit. Placing names inside the aircraft will avoid removal and replacement prior to or after deployments.
COMMAND MINUTEMAN INSIGNIA: 30 inches, (flat black or gray decal) Optional. Both side of vertical stabilizer centered with the flag. The top of the emblem is 10 inches from the bottom of the call number.
STATE NAME: 10 inch (37038), Centered on the main landing gear pod horizontally and centered vertically as the APU exhaust will allow. Vertical placement should be symmetrical for both pods.

Attachment 4

DAILY SAS REPORTING METRIC

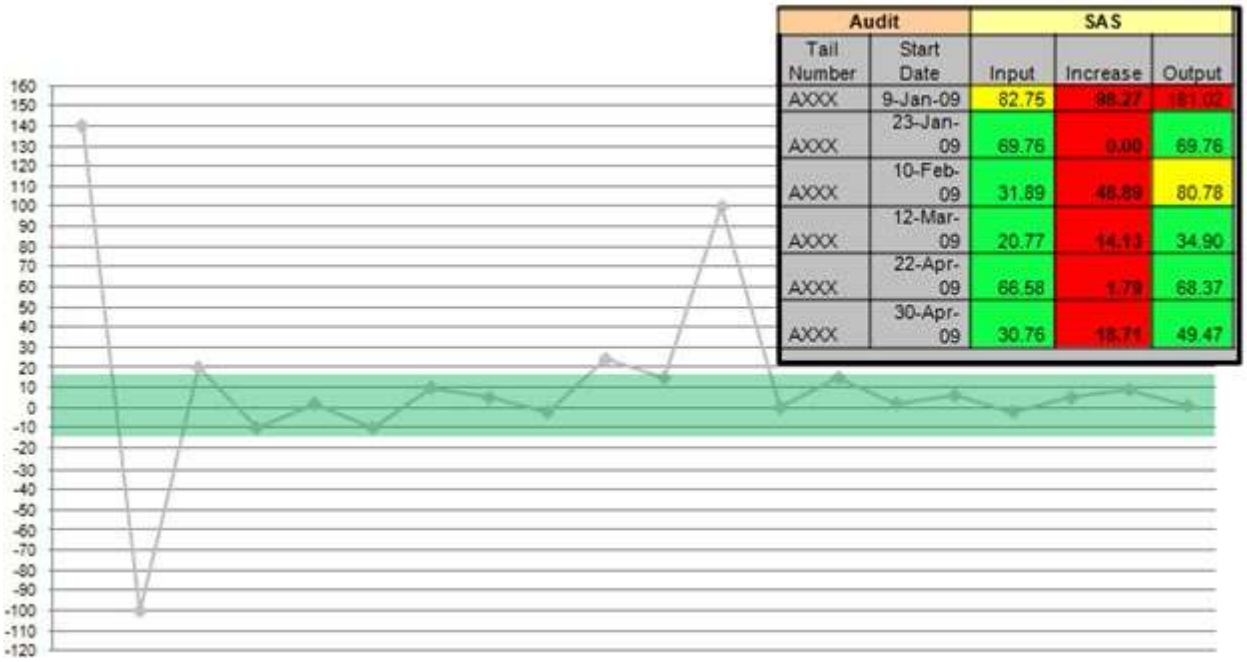
Figure A4.1. Daily SAS Reporting Metric.



Attachment 5

SAS AUDIT METRIC

Figure A5.1. SAS Audit Metric.



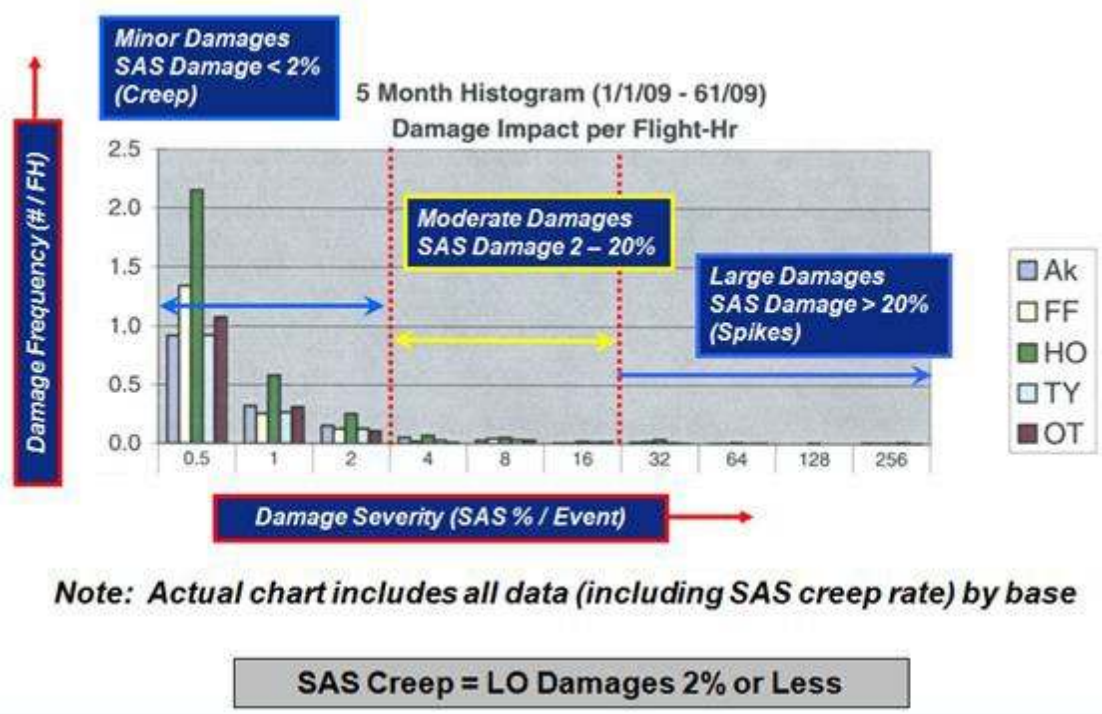
Audit		SAS		
Tail Number	Start Date	Input	Increase	Output
AXXX	9-Jan-09	82.75	98.27	181.02
AXXX	23-Jan-09	69.76	0.00	69.76
AXXX	10-Feb-09	31.89	48.89	80.78
AXXX	12-Mar-09	20.77	14.13	34.90
AXXX	22-Apr-09	66.58	1.79	68.37
AXXX	30-Apr-09	30.76	18.71	49.47

Attachment 6

SAS DAMAGE IMPACT DEFINITION

A6.1. The chart below shows a categorization approach to optimize maintenance activities. It outlines three categories of damages, those with an impact greater than 20% (major "spikes"), those between 2% and 20% (moderate "routine") and those less than 2% (minor "creep"). Post flight OML inspections accomplished by the crew chief and/or LO personnel must identify any LO spike damages as soon as possible after flight if the aircraft is on the next day flying schedule.

Figure A6.1. SAS DAMAGE IMPACT CHART.



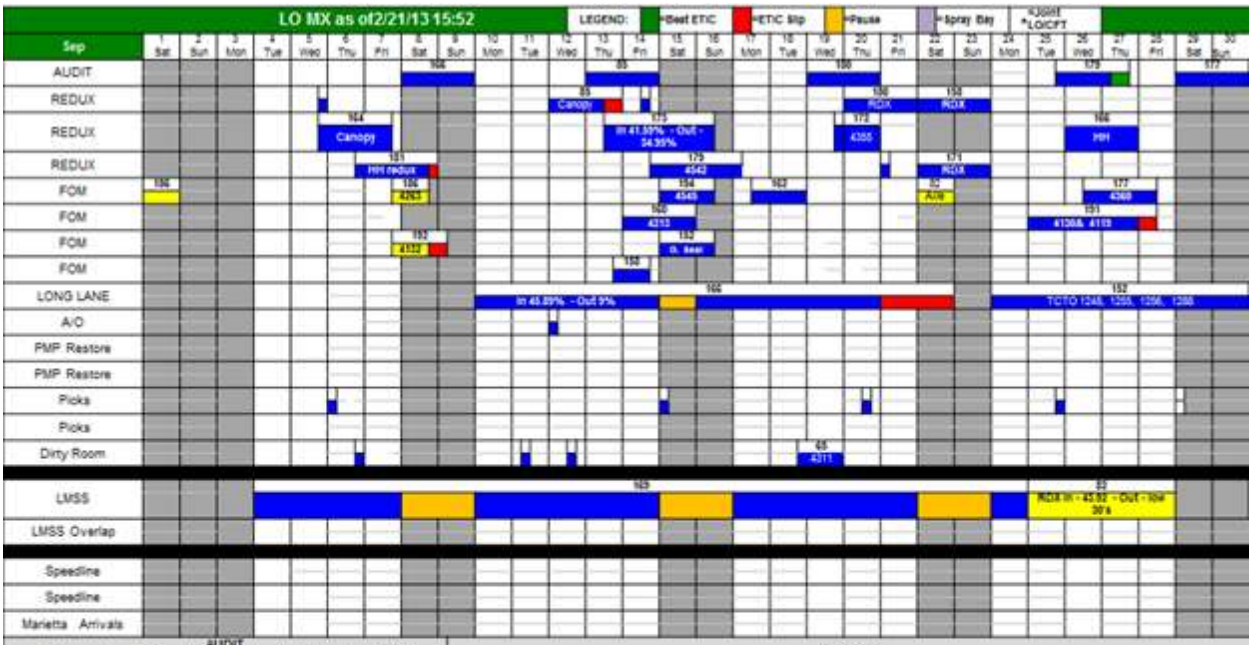
A6.1.1. The damage definition/SAS creep metric must be used to establish a battle rhythm for managing LO maintenance. Minor damages are repaired through scheduled SAS redux. Moderate damages should primarily be worked in groups in conjunction with other scheduled maintenance, e.g., Packaged Maintenance Plans, TCTOs, and panel removals to facilitate other maintenance. Any existing moderate damages should be the priority when performing scheduled SAS reduction efforts. Large damages that drive significant increases in SAS must be fixed as soon as possible to manage SAS growth. In some cases, it may be prudent to define large damages as >10 percent to effectively control SAS margins. This more aggressive approach is particularly beneficial prior to major deployments. Units have the option to define spikes as >10% if necessary to control spike growth, but the SAS creep definition provided above must be used in all cases.

A6.1.2. Use wing analysis and scheduling experts to help balance flying operations and LO maintenance events/downtime to best manage LO fleet health. Failure to effectively balance flying and LO maintenance requirements could lead to an uncontrollable LO backlog.

Attachment 7

LO MAINTENANCE SCHEDULE

Figure A7.1. LO Maintenance Schedule.



AUDIT				
A/C	LOC	SAS	Started	ETIC
179	180	31.26	25-Sep-12	26 sep 12

Remarks

REDUX				
A/C	LOC	SAS	Started	ETIC
153	LO2	42.27	24 Sep 0700	5 Oct 2359
166	180	103.08	25 Sep 0430	28 Sep 0430

Remarks

FOM				
A/C	LOC	SAS	Started	ETIC
191	180	21.91	25 Sep 12	28 Sep 0900
177	180	23.92	27 Sep 1130	29 Sep 1130

4360

Remarks

Dedicated CFT (LMSS)				
A/C	LOC	SAS	Started	ETIC
82	LO3	43.92	25 Sep 0700	28 Sep 1600

Remarks

AW LO				
A/C	N/A	N/A	N/A	N/A

Remarks

Attachment 8

CANOPY TRANSPARENCY STATUS REPORTING.

A8.1. Canopy transparency coating data is tracked by AF Engineering Technical Services, Field Service Representatives or Contract Engineering Technical Services in a format similar to the chart below. The tracking sheet must also include transparency manufacture and damage information by placing a flag note in each block used to track canopy hours. This data is tracked to help units predict reliability. Canopy data must be disseminated to MXG supervisors, MAJCOM weapon system team and the LM canopy IPT at least weekly.

Figure A8.1. Canopy Transparency Status Chart.

A/C	CANOPY HOURS	Visual Status	A/C	CANOPY HOURS	Visual Status	A/C	CANOPY HOURS	Visual Status	A/C	CANOPY HOURS	Visual Status
05-090	260.7	8	06-118	154.3	9	07-131	92.7		07-142	193	20
05-102	95.7		06-119	8.5	1	07-133	49.4		07-143	143	4
05-103	3.6		06-121	184.5	6	07-134	62.4		07-144	147.9	3
06-108	273	3	06-122	397		07-135	158		07-145	108.7	
06-110	73.5	2	06-123	192.9	12	07-136	85		07-146	214.1	6
06-112	76.3					07-137	23.8		07-147	293.1	15
06-113	189.1	4	06-126	148.7	2	07-138	179.4	2	07-148	283.8	10
06-114	380.3	10	06-127	112	5	07-139	75.7		07-149	139.8	2
06-115	152.6		06-129	114.5		07-140	167.7	25	07-150	96.7	2
06-117	111.3	1	06-130	58.8	1	07-141	119.4	6	07-151	173.7	31

Canopy Hours	Visual Status
1-99.9 Hours	Major Visual Anomalies
100 - 199.9 Hours	Minor Visual Anomalies
200+ - Hours	No Visual Anomalies
	Number Inserted In Box = Repairs In Zone 1